

EXACT COUNTS OF ELLIPTIC CURVES OF BOUNDED HEIGHT OVER $\mathbb{F}_q(t)$ IN CHARACTERISTICS 2 AND 3

JUN-YONG PARK

ABSTRACT. Let $p \in \{2, 3\}$, let $q = p^r$ with $r \geq 1$, and put $K = \mathbb{F}_q(t)$. We determine the exact number of K -isomorphism classes of elliptic curves of bounded Faltings height, equivalently of bounded minimal-discriminant degree.

Two small-characteristic phenomena enter the count. First, certain generalized Weierstrass equations with nonsmooth generic fiber have a unique geometric singular point defined only after a nontrivial purely inseparable extension of K . Second, the extra K -defined automorphisms on the $j = 0$ locus, including wild automorphisms, must be incorporated when passing from weighted to unweighted counts. Building on de Jong's weighted-counting framework, we correct the nonsmooth-locus subtraction, classify normalized fixed pairs with a marked automorphism, and transfer the resulting coefficient-degree counts to exact Faltings height through the intrinsic effective divisor recording minimality defect. This yields closed unweighted formulas and identifies the geometric origin of every lower-order term. At height zero, the $j = 0$ contribution agrees with the finite-field twist counts of Kronberg–Soomro–Top.

1. INTRODUCTION

Let $p \in \{2, 3\}$, let $q = p^r$ with $r \geq 1$, and put

$$K = \mathbb{F}_q(t).$$

Our aim is to determine, for every height bound, the unweighted number of K -isomorphism classes of elliptic curves over K . For finite fields of characteristic greater than 3, the corresponding formulas were obtained in [BPS22, Theorem 9.7]. Characteristics 2 and 3 are not formal specializations of that calculation: the nonsmooth locus contains additional purely inseparable strata, and the passage from weighted to unweighted counts must account for the extra automorphisms on the $j = 0$ locus.

For an elliptic curve E/K , let

$$f : X \longrightarrow \mathbb{P}_{\mathbb{F}_q}^1$$

be its relatively-minimal proper regular model, and let $\sigma : \mathbb{P}_{\mathbb{F}_q}^1 \rightarrow X$ be the section extending the identity point O . Write $\omega_{X/\mathbb{P}^1}$ for the relative dualizing sheaf. Its fundamental line bundle is

$$\mathcal{L}_f = (R^1 f_* \mathcal{O}_X)^\vee \simeq f_* \omega_{X/\mathbb{P}^1},$$

and its Faltings height is the nonnegative integer

$$\text{ht}(E) = \deg \mathcal{L}_f.$$

The minimal discriminant divisor satisfies

$$\deg \Delta_{\min}(E) = 12 \text{ht}(E).$$

Throughout, isomorphisms and automorphisms of elliptic curves preserve O . Isomorphisms of elliptic surfaces lie over the identity of $\mathbb{P}_{\mathbb{F}_q}^1$ and preserve the zero section. We call an elliptic curve E/K *constant* if it is K -isomorphic to the base change of an elliptic curve $E_0/\mathbb{F}_q$. We call E *isotrivial* if there exist a finite separable extension L/K and an elliptic curve $E_0/\mathbb{F}_q$ such that

$$E_L \simeq E_0 \times_{\mathbb{F}_q} L.$$

A generalized Weierstrass equation has *coefficient degree* d when its Weierstrass line bundle is $\mathcal{O}(d)$, so that

$$a_i \in H^0(\mathbb{P}^1, \mathcal{O}(id)), \quad i \in \{1, 2, 3, 4, 6\}.$$

Such an equation need not be relatively-minimal. By its *resolved model* we mean the minimal desingularization of the associated Weierstrass surface. Existence of a desingularization in every characteristic follows from resolution of singularities for excellent surfaces [Lip78]. If its generic fiber is smooth of exact Faltings height n , de Jong's construction associates to its resolved model an effective blowup divisor D on $\mathbb{P}^1$ satisfying

$$d = n + \deg D.$$

By analogy with [BPS22, Definition 4.31], we call

$$e = d - n = \deg D$$

the *minimality defect*. The divisor D plays the role of the effective quotient divisor in [BPS22, Lemma 4.32].

The starting point is de Jong's weighted-counting framework, in which a class $[E/K]$ has weight

$$\frac{1}{|\mathrm{Aut}_K(E, O)|}.$$

The rational-singular-section parametrization in [dJ02, Section 4.11] contributes q^{8d+4} coefficient-degree- d equations whose generic fiber has a K -rational singular point. Over the imperfect field K , this does not exhaust the nonsmooth locus in characteristics 2 and 3: certain equations have a unique geometric singular point defined only after a nontrivial purely inseparable extension of K . Its closure over $\mathbb{P}^1$ is then a multisection rather than a section. Lemma 2.3 determines the additional contribution, and Proposition 2.5 gives the corrected weighted count.

To obtain the unweighted count, we must also account for K -defined automorphisms. Every elliptic curve has the two distinct central automorphisms 1 and $[-1]$. Over an algebraic closure, automorphisms beyond these occur only when $j = 0$; in that case the origin-preserving automorphism group has order 12 in characteristic three and order 24 in characteristic two [Sil09, Appendix A, Proposition 1.2(c)]. As shown in Remark 2.2, the corresponding automorphism group schemes are nevertheless finite and étale.

We therefore count marked pairs

$$(E, \alpha), \quad \alpha \in \mathrm{Aut}_K(E, O) \setminus \{1, [-1]\}.$$

The automorphism group of (E, α) is the centralizer of α in $\mathrm{Aut}_K(E, O)$, so the marked-pair groupoid supplies exactly the centralizer weights required by inertia.

Although the curves on the $j = 0$ locus have the same geometric automorphism group, twisting changes which automorphisms descend to K .

At the level of normalized equations, the admissible coordinate changes preserving the normalized form constitute the *residual coordinate group*. A *fixed pair* consists of a normalized coefficient tuple together with an element of this group that fixes it. Some of the resulting geometric fixed-pair loci are individually defined over $\mathbb{F}_q$ exactly when r is even.

These fixed-pair calculations initially count equations of coefficient degree d . We retain the marked automorphism while passing to the relatively-minimal model and then invert the convolution over the possible effective divisors D . The intrinsic divisor is essential: visible coefficient divisibility is not preserved by the residual coordinate translations in characteristics 2 and 3.

For $q = p^r$, set

$$(1) \quad \epsilon_r = \begin{cases} 1, & r \text{ even,} \\ 0, & r \text{ odd.} \end{cases}$$

For every integer $m \geq 2$, put

$$(2) \quad \alpha_m(q) = \sum_{i=0}^{m-2} q^{-i} = \frac{q^{m-1} - 1}{q^{m-2}(q-1)}, \quad \beta_m(q) = \sum_{i=1}^{m-1} q^{-i} = \frac{q^{m-1} - 1}{q^{m-1}(q-1)} = \frac{\alpha_m(q)}{q}.$$

Define the multiplicative discriminant height by

$$H_\Delta(E) = q^{\deg \Delta_{\min}(E)} = q^{12\text{ht}(E)}.$$

For every real number $B \geq 1$, put

$$N_{q,\Delta}(B) = \#\{[E/K] : H_\Delta(E) \leq B\}.$$

Theorem 1.1. *Let $p \in \{2, 3\}$ and $r \geq 1$, and put $q = p^r$. For the positive-height formulas, let*

$$B = q^{12M}, \quad M \in \mathbb{Z}_{\geq 1}.$$

If $p = 3$, then

$$\begin{aligned} N_{q,\Delta}(B) &= 2 \frac{q^9 - 1}{q^8 - q^7} B^{5/6} + 2\alpha_6(q) B^{1/2} \\ &\quad + 2(\epsilon_r \alpha_4(q) - 2\beta_4(q)) B^{1/3} - 2B^{1/6} + 2\epsilon_r - 2. \end{aligned}$$

If $p = 2$, then

$$\begin{aligned} N_{q,\Delta}(B) &= 2 \frac{q^9 - 1}{q^8 - q^7} B^{5/6} + 2(2\epsilon_r \alpha_6(q) - \beta_6(q)) B^{1/2} \\ &\quad + \alpha_5(q) B^{5/12} - \beta_4(q) B^{1/3} - 4\epsilon_r \beta_3(q) B^{1/4} - 2B^{1/6}. \end{aligned}$$

For height zero, corresponding to $M = 0$ and $B = 1$,

$$N_{q,\Delta}(1) = \begin{cases} 2q + 2 + 2\epsilon_r, & p = 3, \\ 2q + 1 + 4\epsilon_r, & p = 2. \end{cases}$$

For $B = q^{12M}$, the leading term of order $B^{5/6}$ is twice the leading term of the corrected cumulative weighted count. The lower-order terms have three sources: the corrected subtraction of nonsmooth equations, the marked extra-automorphism

loci, and the removal of the minimality defect. The factor ϵ_r records whether the relevant geometric automorphism loci descend individually to $\mathbb{F}_q$. Section 7 gives the full term-by-term interpretation.

At height zero, the $j = 0$ locus contributes $4 + 2\epsilon_r$ constant K -isomorphism classes in characteristic three and $3 + 4\epsilon_r$ in characteristic two, in agreement with [KST17, Proposition 2.2] and [KST17, Proposition 3.2], respectively.

The proof uses generalized Weierstrass equations rather than the isomorphism

$$\overline{\mathcal{M}}_{1,1} \simeq \mathcal{P}(4,6)$$

over $\mathbb{Z}[1/6]$ [BPS22, p. 3], where $\mathcal{P}(4,6)$ is the quotient of $\mathbb{A}^2 \setminus \{0\}$ by the $\mathbb{G}_m$ -action with weights 4 and 6. This isomorphism does not extend to characteristics 2 or 3. In characteristic 2, the generic stabilizer of the corresponding scalar action is the nonreduced group scheme μ_2 , whereas inversion in the origin-preserving automorphism group generates a constant étale subgroup of order two. In characteristic 3, the wild automorphisms arise from residual translations of the x -coordinate rather than from root-of-unity stabilizers of the scalar action on $\mathcal{P}(4,6)$. Working with resolved Weierstrass models and de Jong's invariant blowup divisor avoids these difficulties. The nonsmooth correction is logically distinct from the wild automorphism calculation. Wildness concerns automorphisms whose orders are divisible by the characteristic, whereas the imperfection of K allows certain geometric singular points to be purely inseparable over K rather than K -rational.

1.1. Outline of the paper. Section 2 corrects the nonsmooth equation count, derives the weighted count, and formulates the marked-inertia correction. Sections 3 and 4 normalize the $j = 0$ equations and classify the extra fixed pairs in characteristics 3 and 2. Section 5 transfers these ambient marked counts from coefficient degree to exact Faltings height using the intrinsic blowup divisor. Section 6 proves the exact counting formulas. Finally, Section 7 compares the height-zero terms with the finite-field twist classification and explains the geometric origins of the lower-order terms.

2. WEIGHTED COUNTS AND MARKED INERTIA

Call a groupoid *essentially finite* if it has finitely many isomorphism classes and every object has a finite automorphism group. For any groupoid $\mathcal{G}$, write

$$\pi_0(\mathcal{G})$$

for the set of isomorphism classes of objects of $\mathcal{G}$. If $\mathcal{G}$ is essentially finite, define its weighted cardinality by

$$\#_q \mathcal{G} = \sum_{[\xi] \in \pi_0(\mathcal{G})} \frac{1}{|\text{Aut}(\xi)|}.$$

For $n \geq 0$, let $\mathcal{E}_n(q)$ be the groupoid of elliptic curves $(E, O)/K$ of Faltings height n , with morphisms given by origin-preserving K -isomorphisms. The finite coefficient spaces used below and Remark 2.2 show that $\mathcal{E}_n(q)$ and the marked groupoids introduced below are essentially finite.

Define

$$I_n(q) = \#_q \mathcal{E}_n(q), \quad U_n(q) = |\pi_0(\mathcal{E}_n(q))|.$$

Thus $I_n(q)$ is the weighted count, while $U_n(q)$ is the number of K -isomorphism classes.

Let $\mathcal{E}_n^{\text{ex}}(q)$ be the groupoid whose objects are pairs (E, α) with

$$E \in \mathcal{E}_n(q), \quad \alpha \in \text{Aut}_K(E, O) \setminus \{1, [-1]\},$$

and whose morphisms

$$\phi : (E, \alpha) \longrightarrow (E', \alpha')$$

are origin-preserving K -isomorphisms satisfying

$$\phi \alpha \phi^{-1} = \alpha'.$$

Set

$$X_n(q) = \#_q \mathcal{E}_n^{\text{ex}}(q).$$

The full inertia groupoid of $\mathcal{E}_n(q)$ has objects (E, α) with

$$E \in \mathcal{E}_n(q), \quad \alpha \in \text{Aut}_K(E, O),$$

and morphisms given by conjugation as above. Thus $\mathcal{E}_n^{\text{ex}}(q)$ is the part obtained by removing the identity and inversion from every automorphism group.

Recall that every origin-preserving K -isomorphism

$$\phi : (E, O) \longrightarrow (E', O')$$

is an isomorphism of group varieties [Sil09, Chapter III, Theorem 4.8]. Consequently,

$$\phi \circ [-1]_E = [-1]_{E'} \circ \phi.$$

In particular, $[-1]$ is central in $\text{Aut}_K(E, O)$, and conjugation by an isomorphism carries $\{1_E\}$, $\{[-1]_E\}$, and their complement to the corresponding subsets for E' .

The following lemma is the finite-groupoid form of the general identity that the weighted point count of the inertia stack equals the unweighted number of rational isomorphism classes [HP23, Theorem 1.1]. We retain the direct proof because it explicitly separates the identity, inversion, and extra-automorphism parts and records the corresponding centralizer weights.

Lemma 2.1. *For every $n \geq 0$,*

$$U_n(q) = 2I_n(q) + X_n(q).$$

Proof. Fix an isomorphism class $[E]$ and put

$$A_E = \text{Aut}_K(E, O).$$

The groupoid of pairs (E', α) with $E' \simeq E$ and

$$\alpha \in \text{Aut}_K(E', O')$$

is equivalent to the conjugation action groupoid

$$A_E // A_E.$$

Here the first copy of A_E is the set of automorphisms being marked, and the second copy acts on it by conjugation:

$$g \cdot \alpha = g \alpha g^{-1}.$$

Thus the isomorphism classes of $A_E // A_E$ are the conjugacy classes in A_E . The automorphism group of the object α is its centralizer

$$C_{A_E}(\alpha) = \{g \in A_E : g\alpha = \alpha g\}.$$

By the orbit-stabilizer formula,

$$|[\alpha]_{\text{conj}}| = \frac{|A_E|}{|C_{A_E}(\alpha)|}.$$

Consequently,

$$\begin{aligned} \#_q A_E // A_E &= \sum_{[\alpha]_{\text{conj}} \subset A_E} \frac{1}{|C_{A_E}(\alpha)|} \\ &= \frac{1}{|A_E|} \sum_{[\alpha]_{\text{conj}} \subset A_E} |[\alpha]_{\text{conj}}| \\ &= 1, \end{aligned}$$

because the conjugacy classes partition A_E . The identity and inversion are distinct central elements of A_E , including in characteristic two as explained in Remark 2.2. Hence their two conjugacy classes contribute

$$\frac{2}{|A_E|}.$$

The remaining conjugacy classes contribute

$$1 - \frac{2}{|A_E|} = \frac{|A_E| - 2}{|A_E|}.$$

These remaining conjugacy classes are precisely the isomorphism classes in the fiber of $\mathcal{E}_n^{\text{ex}}(q)$ over $[E]$, and the automorphism group of (E, α) is

$$C_{A_E}(\alpha).$$

Summing over $[E] \in \pi_0(\mathcal{E}_n(q))$ proves the result. ■

Remark 2.2. Let $\text{char}(K) \in \{2, 3\}$. The automorphism functor is represented by a finite-type group scheme

$$G = \underline{\text{Aut}}_K(E, O).$$

For a K -algebra R , its R -points are

$$G(R) = \text{Aut}_R(E_R, O_R), \quad E_R = E \times_K \text{Spec } R.$$

In particular,

$$G(K) = \text{Aut}_K(E, O).$$

The Lie algebra of G is its tangent space at the identity:

$$\text{Lie}(G) = T_1 G = \ker \left(G(K[\varepsilon]/(\varepsilon^2)) \longrightarrow G(K) \right).$$

Thus $\text{Lie}(G)$ parametrizes first-order automorphisms that reduce to the identity when $\varepsilon = 0$.

First-order automorphisms of E correspond to global vector fields on E . Requiring the automorphism to fix O forces the vector field to vanish at O . Therefore,

$$\text{Lie}(G) = H^0(E, T_E(-O)).$$

An invariant differential trivializes the cotangent bundle of an elliptic curve, so

$$T_E \simeq \mathcal{O}_E.$$

It follows that

$$\mathrm{Lie}(G) = H^0(E, \mathcal{O}_E(-O)) = 0,$$

because $\mathcal{O}_E(-O)$ has degree -1 . Hence G is unramified at the identity. After base change to an algebraic closure of K , left translation in the group scheme identifies the tangent space at every geometric point with the tangent space at the identity, so it is unramified everywhere. Since every scheme over a field is flat, and finite type over a field implies finite presentation, $\underline{\mathrm{Aut}}_K(E, O)$ is étale over K .

An étale K -scheme of finite type is a finite disjoint union of spectra of finite separable extensions of K . Consequently, $\underline{\mathrm{Aut}}_K(E, O)$ is finite étale.

In characteristic two, although the field elements -1 and 1 coincide, the inversion morphism

$$[-1]: E \longrightarrow E$$

is not the identity morphism. For every elliptic curve and every nonzero integer m , the multiplication morphism

$$[m]: E \longrightarrow E$$

is a finite isogeny of degree m^2 ; equivalently, its kernel group scheme $E[m]$ has rank m^2 . Hence

$$\deg[2] = 4$$

even in characteristic two.

If $[-1]$ were the identity, then

$$[2](P) = P + P = P + (-P) = O$$

for every P , so $[2]$ would be the constant zero morphism. This contradicts the fact that $[2]$ is a finite morphism of degree four.

For $j(E) \neq 0$, the geometric automorphism group consists of 1 and $[-1]$ [Sil09, Appendix A, Proposition 1.2(c)]. Since both automorphisms are defined over K , the automorphism group scheme is the constant group scheme with these two elements. All weighted counts below use the finite group

$$\mathrm{Aut}_K(E, O)$$

of K -defined automorphisms.

We next revisit the geometric setup underlying de Jong's weighted count and refine its nonsmooth-locus calculation in characteristics 2 and 3. We use d for the degree of the line bundle in which a possibly nonminimal Weierstrass equation is written. Thus a coefficient-degree- d equation has

$$a_i \in H^0(\mathbb{P}^1, \mathcal{O}(id)), \quad i = 1, 2, 3, 4, 6.$$

The integer d is not the Faltings height unless the resulting model is relatively-minimal. Thus n denotes exact Faltings height, d denotes coefficient degree, and M is reserved for the cutoff in cumulative counts.

If a coefficient-degree- d equation has a relatively-minimal model of height n , then de Jong's effective blowup divisor has degree

$$e = d - n.$$

By analogy with [BPS22, Definition 4.31], we call e the *minimality defect*. It does not change the generic elliptic curve; it records the nonminimality of the chosen Weierstrass model over $\mathbb{P}^1$.

Among the q^{16d+5} coefficient-degree- d equations, de Jong's singular-section argument counts q^{8d+4} equations whose generic fiber has a K -rational singular point [dJ02, Section 4.11]. Because $K = \mathbb{F}_q(t)$ is imperfect, this need not exhaust the nonsmooth locus: the unique geometric singular point may be defined only over a purely inseparable extension. As the proof below shows, the additional equations can occur only in the locus $a_1 = a_3 = 0$ in characteristic two or, after completing the square, in the locus $a_2 = a_4 = 0$ in characteristic three. We call these the *quasi-elliptic-type strata*. The following lemma gives the full count.

Lemma 2.3. *Let p be a prime, let $r \geq 1$, put $q = p^r$, and let $d \geq 0$. If $p \in \{2, 3\}$, the number of coefficient-degree- d generalized Weierstrass equations over $\mathbb{P}_{\mathbb{F}_q}^1$ whose generic fiber is not smooth equals*

$$\begin{cases} q^{8d+4} + q^{10d+3} - q^{6d+3}, & p = 3, \\ q^{8d+4} + q^{12d+3} - q^{7d+3}, & p = 2. \end{cases}$$

If $p \geq 5$, the number is q^{8d+4} . In particular, for $d = 0$ the answer is q^4 in every characteristic.

Proof. Write the equation as $F = 0$, where

$$F = y^2 + a_1xy + a_3y - x^3 - a_2x^2 - a_4x - a_6, \quad a_i \in H^0(\mathbb{P}^1, \mathcal{O}(id)).$$

The generic fiber is smooth if and only if its discriminant is nonzero in K .

We first count the equations whose generic fiber has a K -rational singular point. After homogenization, the generic Weierstrass cubic meets the line at infinity only at

$$O = [0 : 1 : 0].$$

This point is smooth, since the derivative of the homogenized equation with respect to the homogenizing variable equals 1 at O . Every positive-degree projective component meets the line at infinity. Since the cubic meets that line only at O , every geometric component passes through O . Thus, if the cubic were reducible, at least two components would pass through O ; if it were nonreduced, a repeated component would pass through O . Either possibility would make O singular. Thus the cubic is geometrically integral. An integral plane cubic has at most one geometric singular point.

The closure of a K -rational singular point is, by properness, a section of the associated Weierstrass surface contained in the relative singular locus. The same derivative calculation shows that the zero section lies in the relative smooth locus, and every fiber meets the line at infinity only in the zero section. Hence the singular section lies in the affine chart. Writing (x, y) for the affine coordinates, it has the form

$$(x, y) = (C_2, C_3), \quad C_2 \in H^0(\mathbb{P}^1, \mathcal{O}(2d)), \quad C_3 \in H^0(\mathbb{P}^1, \mathcal{O}(3d)).$$

At (C_2, C_3) , the equations $F = F_x = F_y = 0$ are equivalent to

$$\begin{aligned} a_3 &= -2C_3 - a_1C_2, \\ a_4 &= a_1C_3 - 3C_2^2 - 2a_2C_2, \\ a_6 &= -C_3^2 - a_1C_2C_3 + 2C_2^3 + a_2C_2^2. \end{aligned}$$

Hence (a_1, a_2, C_2, C_3) parametrizes these equations bijectively, since the geometric singular point is unique. Thus exactly

$$q^{(d+1)+(2d+1)+(2d+1)+(3d+1)} = q^{8d+4}$$

equations have a K -rational singular point, in every characteristic.

Suppose first that $p = 2$. On the affine chart,

$$\frac{\partial F}{\partial y} = a_1x + a_3.$$

If $a_1 \neq 0$, every geometric singular point satisfies

$$x_0 = \frac{a_3}{a_1}, \quad a_1y_0 = x_0^2 + a_4,$$

and is therefore K -rational. If $a_1 = 0$ and $a_3 \neq 0$, then $\partial F / \partial y = a_3$ is nonzero on the generic fiber, so the generic fiber is smooth. If $a_1 = a_3 = 0$, the singularity equations over $\bar{K}$ are

$$x_0^2 = a_4, \quad y_0^2 = a_2a_4 + a_6.$$

They always have a solution, so all

$$q^{(2d+1)+(4d+1)+(6d+1)} = q^{12d+3}$$

equations in this stratum are nonsmooth.

Within this stratum, the singular point is K -rational precisely when there exist $C_2, C_3 \in K$ such that

$$a_4 = C_2^2, \quad a_2a_4 + a_6 = C_3^2.$$

Taking valuations in these identities, using local trivializations at every closed point, shows that any such C_2 and C_3 extend to global sections of the indicated line bundles. Consequently, we may equivalently require

$$C_2 \in H^0(\mathbb{P}^1, \mathcal{O}(2d)), \quad C_3 \in H^0(\mathbb{P}^1, \mathcal{O}(3d)).$$

Frobenius is injective on these spaces of sections, so the triples (a_2, C_2, C_3) parametrize exactly

$$q^{(2d+1)+(2d+1)+(3d+1)} = q^{7d+3}$$

members of this stratum already counted by the rational-singular parametrization. Inclusion–exclusion therefore gives

$$q^{8d+4} + q^{12d+3} - q^{7d+3}.$$

Now suppose that $p = 3$. Completing the square removes a_1 and a_3 . For each fixed pair (a_1, a_3) , this is an affine-linear bijection on the remaining coefficient space, and there are

$$q^{(d+1)+(3d+1)} = q^{4d+2}$$

choices of (a_1, a_3) . Renaming the transformed coefficients a_2, a_4, a_6 , the completed-square equation is

$$y^2 = f(x), \quad f = x^3 + a_2x^2 + a_4x + a_6.$$

A singular point satisfies

$$y_0 = 0, \quad f(x_0) = f'(x_0) = 0, \quad f' = 2a_2x + a_4.$$

If $a_2 \neq 0$, every repeated root is

$$x_0 = -\frac{a_4}{2a_2} \in K.$$

If $a_2 = 0$ and $a_4 \neq 0$, then $f' = a_4$ is nonzero, so the generic fiber is smooth. If $a_2 = a_4 = 0$, then

$$f = x^3 + a_6.$$

For each fixed pair (a_1, a_3) , all q^{6d+1} choices of a_6 in this stratum are nonsmooth, and their singular point is K -rational precisely when

$$a_6 = -C_2^3$$

for some $C_2 \in K$. Taking valuations in this identity, using local trivializations at every closed point, shows that any such C_2 extends to a section of $\mathcal{O}(2d)$. Thus

$$C_2 \in H^0(\mathbb{P}^1, \mathcal{O}(2d)),$$

and there are q^{2d+1} such equations.

For a rational singular point with $x = C_2$, the equations $f'(C_2) = f(C_2) = 0$ give

$$a_4 = -2a_2C_2, \quad a_6 = -C_2^3 + a_2C_2^2.$$

Thus the free pair (a_2, C_2) gives q^{4d+2} equations. Inclusion–exclusion in that space, followed by restoring the q^{4d+2} choices of (a_1, a_3) , gives

$$q^{4d+2}(q^{4d+2} + q^{6d+1} - q^{2d+1}) = q^{8d+4} + q^{10d+3} - q^{6d+3}.$$

For $p \geq 5$, complete the square and depress the cubic, obtaining

$$y^2 = f(x), \quad f = x^3 + Ax + B.$$

These coordinate changes are defined over K and preserve the rationality of singular points. A singular point has $y = 0$ and is a common root of f and

$$f' = 3x^2 + A.$$

If $A = 0$, the only root of f' is $0 \in K$. If $A \neq 0$, then f' is separable. An irrational common root of f and f' would have a distinct conjugate that was also a common root, giving two distinct repeated roots of a cubic, which is impossible. Hence every nonsmooth equation has a K -rational singular point, and its number is q^{8d+4} .

Finally, when $d = 0$, Frobenius is bijective on $\mathbb{F}_q$, so the small-characteristic strata add no equations beyond the rational-singular locus. Both displayed formulas reduce to q^4 . ■

Remark 2.4. The step in [dJ02, Section 4.11] that requires modification is the assertion that the unique geometric singular point extends to a section over $\mathbb{P}^1$. The resulting q^{8d+4} parameter count therefore counts exactly the equations whose singular point is K -rational. Over the imperfect field $K = \mathbb{F}_q(t)$, geometric uniqueness does not imply K -rationality: the point may be defined only over a purely inseparable extension.

For example, $y^2 = x^3 + t$ has geometric singular point $(0, t^{1/2})$ in characteristic two and $(-t^{1/3}, 0)$ in characteristic three. Since $t \notin K^p$ —a p -th power has valuation divisible by p —this point is not K -rational. The normalization of its closure maps purely inseparably to $\mathbb{P}^1$ with degree p , rather than giving a section.

Consequently, in characteristics two and three and for $d \geq 1$, the term q^{8d+4} used for the type-(b) locus in the proof of [dJ02, Proposition 4.12] must be replaced by the corresponding formula in Lemma 2.3. This correction affects only the type-(b) nonsmooth term; the type-(a) blowup-divisor construction in [dJ02, Section 4.11] is unchanged. The additional contributions come from the portions of the quasi-elliptic-type strata that do not possess a rational singular section. Quasi-elliptic fibrations occur only in characteristics two and three [BM76]. Thus the correction changes lower-order terms but not the leading weighted asymptotic.

Let $W_d^{\text{sm}}(q)$ be the finite set of coefficient-degree- d generalized Weierstrass equations whose generic fiber is smooth. Let $G_d(q)$ be the group of global Weierstrass coordinate changes acting on $W_d^{\text{sm}}(q)$. Explicitly, these changes have the form

$$x = u^2 x' + \rho, \quad y = u^3 y' + u^2 \eta x' + \tau,$$

where

$$u \in \mathbb{F}_q^\times, \quad \rho \in H^0(\mathbb{P}^1, \mathcal{O}(2d)), \quad \eta \in H^0(\mathbb{P}^1, \mathcal{O}(d)), \quad \tau \in H^0(\mathbb{P}^1, \mathcal{O}(3d)).$$

Define

$$\mathcal{W}_d^{\text{sm}}(q) = W_d^{\text{sm}}(q) // G_d(q)$$

and

$$R_d(q) = \#_q \mathcal{W}_d^{\text{sm}}(q).$$

Thus $R_d(q)$ is the weighted count of generically smooth coefficient-degree- d equations modulo global coordinate change, before imposing relative minimality. For this finite action groupoid,

$$R_d(q) = \frac{|W_d^{\text{sm}}(q)|}{|G_d(q)|}.$$

The group $G_d(q)$ has $(q-1)q^{6d+3}$ elements [dJ02, Sections 4.8–4.9]. Hence Lemma 2.3 gives

$$(3) \quad R_d(q) = \begin{cases} \frac{q^{10d+2} - q^{4d} - q^{2d+1} + 1}{q-1}, & p = 3, \\ \frac{q^{10d+2} - q^{6d} - q^{2d+1} + q^d}{q-1}, & p = 2. \end{cases}$$

In both characteristics,

$$R_0(q) = q.$$

For $e \geq 0$, put

$$c_e(q) = |\mathrm{Sym}^e(\mathbb{P}^1)(\mathbb{F}_q)| = \frac{q^{e+1} - 1}{q - 1}.$$

Here $\mathrm{Sym}^e(\mathbb{P}^1)(\mathbb{F}_q)$ parametrizes effective $\mathbb{F}_q$ -rational divisors of degree e , possibly supported at nonrational closed points. The formula follows from

$$\mathrm{Sym}^e(\mathbb{P}^1) \simeq \mathbb{P}^e.$$

The blowup construction of [dJ02, Section 4.11] associates to a relatively-minimal pair (X, σ) of exact height $n \leq d$ and an effective divisor

$$D = \sum_{x \in |\mathbb{P}^1|} e_x x \in \mathrm{Sym}^{d-n}(\mathbb{P}^1)(\mathbb{F}_q)$$

a coefficient-degree- d model. For the fixed pair (X, σ) , the resulting isomorphism class is determined by D . Here (X, σ) is the relatively-minimal proper regular elliptic surface with zero section associated with an object of $\mathcal{E}_n(q)$; restriction to the generic fiber preserves its origin-preserving automorphism group. Conversely, every generically smooth coefficient-degree- d model arises from such a pair and a divisor D , and relative minimalization recovers both uniquely. The lifting-and-contraction argument of Lemma 5.2 shows that the automorphism group is preserved. Together with the stabilizer-weighted coordinate count of [dJ02, Lemma 4.9], this gives, at the level of weighted cardinalities,

$$(4) \quad R_d(q) = \sum_{n=0}^d c_{d-n}(q) I_n(q).$$

This is the finite-field weighted-point realization of the minimality-defect stratification of [BPS22, Proposition 6.1 and Corollary 6.2]; the corresponding motivic generating-series factorization is [BPS22, Lemma 8.6, equation (6)].

Proposition 2.5. *For every $M \geq 1$,*

$$\sum_{n=0}^M I_n(q) = \begin{cases} \frac{q^9 - 1}{q^8 - q^7} q^{10M} - q^{2M} - \beta_4(q) q^{4M} - 1, & p = 3, \\ \frac{q^9 - 1}{q^8 - q^7} q^{10M} - q^{2M} - \beta_6(q) q^{6M}, & p = 2. \end{cases}$$

At height zero, in both characteristics,

$$I_0(q) = q.$$

Proof. Put

$$R(t) = \sum_{d \geq 0} R_d(q) t^d, \quad I(t) = \sum_{n \geq 0} I_n(q) t^n.$$

Since

$$\sum_{e \geq 0} c_e(q) t^e = \frac{1}{(1-t)(1-qt)},$$

equation (4) gives

$$R(t) = \frac{I(t)}{(1-t)(1-qt)}.$$

For a formal power series $F(t)$, write $[t^M]F(t)$ for the coefficient of t^M . Consequently, for $M \geq 1$,

$$\sum_{n=0}^M I_n(q) = [t^M] \frac{I(t)}{1-t} = [t^M](1-qt)R(t) = R_M(q) - qR_{M-1}(q).$$

Substituting (3) gives, in characteristic three,

$$R_M(q) - qR_{M-1}(q) = \frac{q^9 - 1}{q^8 - q^7} q^{10M} - q^{2M} - \beta_4(q)q^{4M} - 1,$$

and, in characteristic two,

$$R_M(q) - qR_{M-1}(q) = \frac{q^9 - 1}{q^8 - q^7} q^{10M} - q^{2M} - \beta_6(q)q^{6M}.$$

The term $-q^{2M}$ comes from rational singular sections. The extra quasi-elliptic-type corrections are

$$-\beta_4(q)q^{4M} - 1 \quad \text{in characteristic three,} \quad -\beta_6(q)q^{6M} \quad \text{in characteristic two.}$$

Finally, setting $d = 0$ in (4) gives

$$I_0(q) = R_0(q) = q.$$

■

Over an algebraic closure in either characteristic two or three, an elliptic curve has automorphisms beyond

$$\{1, [-1]\}$$

only when $j = 0$ [Sil09, Appendix A, Proposition 1.2(c)]. Consequently, every object of $\mathcal{E}_n^{\text{ex}}(q)$ lies on the $j = 0$ locus. The next two sections give normalized quotient descriptions of this locus and classify its K -defined extra automorphisms.

It remains to compute $X_n(q)$. This is where characteristics 2 and 3 differ. In each characteristic, we first count coefficient-degree- d equations marked by an extra K -defined automorphism and then remove the minimality defect using the marked-automorphism refinement of (4) established in Section 5.

3. EXTRA AUTOMORPHISMS IN CHARACTERISTIC 3

Assume throughout this section that $q = 3^r$ with $r \geq 1$. For $i, d \geq 0$, write

$$V_i(d) = H^0(\mathbb{P}^1, \mathcal{O}(id)).$$

We use the same notation for the associated affine space and, when appropriate, for its additive vector group.

In characteristic three, every generically smooth $j = 0$ Weierstrass equation of coefficient degree d admits a normalized form

$$(5) \quad E_{A,C}: y^2 = x^3 + Ax + C, \quad 0 \neq A \in V_4(d), \quad C \in V_6(d)$$

[dJ02, Section 4.13(b2)]. The condition $A \neq 0$ is equivalent to generic smoothness, since the discriminant is a nonzero scalar multiple of A^3 .

The *residual coordinate group*, namely the group of global Weierstrass coordinate changes remaining after normalization and preserving the form (5), is

$$G_d^{(3)} = \mathbb{G}_m \ltimes V_2(d),$$

where $u \in \mathbb{G}_m$ acts on $V_2(d)$ by multiplication by u^2 . Thus

$$(6) \quad |G_d^{(3)}(\mathbb{F}_q)| = (q-1)q^{2d+1},$$

and the group law is

$$(7) \quad (u_1, s_1)(u_2, s_2) = (u_1 u_2, u_1^2 s_2 + s_1).$$

Its elements act on coordinates by

$$(8) \quad (x, y) \mapsto (u^2 x + s, u^3 y),$$

and hence on normalized coefficients by

$$(9) \quad (u, s) \cdot (A, C) = (u^4 A, u^6 C - s^3 - u^4 A s).$$

Put

$$\mathcal{U}_d^{(3),0} = (V_4(d) \setminus \{0\}) \times V_6(d).$$

For an algebraic group G over $\mathbb{F}_q$, let $H^1(\mathbb{F}_q, G)$ denote the pointed set of isomorphism classes of G -torsors over $\mathbb{F}_q$, with the trivial torsor as distinguished point. A (right) G -torsor over $\mathbb{F}_q$ is an $\mathbb{F}_q$ -scheme P with a right G -action such that, after an fppf (faithfully flat and locally of finite presentation) base change $S \rightarrow \text{Spec } \mathbb{F}_q$, there is a G_S -equivariant isomorphism

$$P_S \simeq G_S,$$

where G_S acts on itself by right translation.

For a left G -scheme X , an object of $[X/G](\mathbb{F}_q)$ is a right G -torsor P together with a morphism

$$f : P \longrightarrow X$$

satisfying

$$f(pg) = g^{-1} \cdot f(p).$$

The natural functor

$$X(\mathbb{F}_q) // G(\mathbb{F}_q) \longrightarrow [X/G](\mathbb{F}_q)$$

selects, up to isomorphism, precisely the objects with trivial underlying torsor. Hence it is an equivalence whenever $H^1(\mathbb{F}_q, G) = \{1\}$.

Proposition 3.1. *Let $\mathcal{W}_d^{(3),0}(q)$ be the groupoid of generically smooth $j = 0$ Weierstrass equations over the fixed base $\mathbb{P}_{\mathbb{F}_q}^1$ of coefficient degree d , with morphisms given by global Weierstrass coordinate changes over the identity of $\mathbb{P}_{\mathbb{F}_q}^1$. Normalization induces an equivalence*

$$\mathcal{W}_d^{(3),0}(q) \simeq [\mathcal{U}_d^{(3),0}/G_d^{(3)}](\mathbb{F}_q).$$

Let $\overline{X}_{A,C} \rightarrow \mathbb{P}^1$ be the corresponding Weierstrass surface and let

$$\tilde{X}_{A,C} \longrightarrow \overline{X}_{A,C}$$

be its minimal desingularization, with lifted zero section $\tilde{\sigma}_{A,C}$. This model need not be relatively-minimal. Then

$$\text{Stab}_{G_d^{(3)}(\mathbb{F}_q)}(A, C) \simeq \text{Aut}_{\mathbb{P}^1}(\tilde{X}_{A,C}, \tilde{\sigma}_{A,C}) \simeq \text{Aut}_K(E_{A,C}, O).$$

Moreover, $G_d^{(3)}$ is smooth and connected, and

$$H^1(\mathbb{F}_q, G_d^{(3)}) = \{1\}.$$

Consequently,

$$[\mathcal{U}_d^{(3),0}/G_d^{(3)}](\mathbb{F}_q) \simeq \mathcal{U}_d^{(3),0}(\mathbb{F}_q) // G_d^{(3)}(\mathbb{F}_q).$$

Proof. de Jong's normalization removes the a_1 - and a_3 -terms, and in the $j = 0$ case the resulting equation has the form (5) [dJ02, Section 4.13(b2)]. A global coordinate change between two normalized equations has the form

$$x \mapsto u^2x + s, \quad y \mapsto u^3y + vx + w,$$

where

$$u \in \mathbb{F}_q^\times, \quad s \in V_2(d), \quad v \in V_1(d), \quad w \in V_3(d).$$

Comparison of the xy - and y -coefficients gives $v = w = 0$, since $2 \neq 0$. The remaining changes are precisely the elements of $G_d^{(3)}(\mathbb{F}_q)$, acting as in (9). We regard (9) as a left coefficient action. This identifies the normalized-equation groupoid with the ordinary action groupoid

$$\mathcal{U}_d^{(3),0}(\mathbb{F}_q) // G_d^{(3)}(\mathbb{F}_q).$$

We prove below that

$$H^1(\mathbb{F}_q, G_d^{(3)}) = \{1\};$$

the preceding description of quotient-stack points then gives the claimed equivalence.

Let $X^{\text{rel}} \rightarrow \mathbb{P}^1$ be the relatively-minimal regular model of the generic fiber. Restriction to the generic fiber gives

$$\text{Aut}_{\mathbb{P}^1}(X^{\text{rel}}, \sigma) \simeq \text{Aut}_K(E_{A,C}, O)$$

[dJ02, Section 4.2]. de Jong's construction obtains $(\tilde{X}_{A,C}, \tilde{\sigma}_{A,C})$ from (X^{rel}, σ) by successively blowing up closed points on the successive strict transforms of the zero section [dJ02, Section 4.11]. Every automorphism fixes the zero section pointwise and hence preserves each blowup center scheme-theoretically. The universal property of the blowup therefore gives a unique lift through every stage of the construction. Conversely, an automorphism of $(\tilde{X}_{A,C}, \tilde{\sigma}_{A,C})$ restricts to an automorphism of the generic fiber. The lift of that restriction agrees with the original automorphism on the generic fiber. Since $\tilde{X}_{A,C}$ is integral and the target is separated, the two morphisms agree everywhere.

Finally, the coordinate-choice construction of [dJ02, Sections 4.8–4.9], applied to the fixed normalized equation, identifies automorphisms of $(\tilde{X}_{A,C}, \tilde{\sigma}_{A,C})$ with global coordinate changes preserving that equation. Since both source and target are normalized, these coordinate changes lie in $G_d^{(3)}(\mathbb{F}_q)$ and are exactly the stabilizer of (A, C) .

As a scheme, the semidirect product $G_d^{(3)}$ is

$$\mathbb{G}_m \times V_2(d).$$

Both factors are smooth and connected, so $G_d^{(3)}$ is smooth and connected. Lang's theorem states that

$$H^1(\mathbb{F}_q, G) = \{1\}$$

for every smooth connected algebraic group G over $\mathbb{F}_q$. Applying it to $G_d^{(3)}$ gives

$$H^1(\mathbb{F}_q, G_d^{(3)}) = \{1\},$$

which proves the final equivalence. ■

For comparison, the weighted cardinality of the entire generically smooth $j = 0$ locus of coefficient degree d is

$$(10) \quad \frac{(q^{4d+1} - 1)q^{6d+1}}{(q - 1)q^{2d+1}} = q^{4d} \frac{q^{4d+1} - 1}{q - 1},$$

which has leading growth q^{8d} . By contrast, the marked extra-automorphism count computed below has smaller leading growth.

Lemma 3.2. *Every origin-preserving K -automorphism of the generic fiber $E_{A,C}$ in (5) has the form (8), with*

$$u \in \mathbb{F}_q^\times, \quad s \in V_2(d),$$

and satisfies

$$(11) \quad u^4 = 1, \quad s^3 + As = (u^6 - 1)C.$$

Apart from the identity $(1, 0)$ and inversion $(-1, 0)$, the solutions are of the following two types.

(i) If $u = \pm 1$ and $s \neq 0$, then

$$A = -s^2,$$

while C is arbitrary. The automorphism $(1, s)$ has order three, whereas $(-1, s)$ has order six.

(ii) If $u^2 = -1$, then

$$A \in V_4(d) \setminus \{0\}, \quad s \in V_2(d)$$

are arbitrary, while

$$C = s^3 + As$$

is forced. The corresponding automorphism has order four. The two roots of $u^2 = -1$ belong to $\mathbb{F}_q$ exactly when r is even.

Proof. By Proposition 3.1, every K -defined automorphism of (5) is given by (8), with

$$u \in \mathbb{F}_q^\times, \quad s \in V_2(d).$$

Substitution into the equation gives (11). Since $A \neq 0$, the equality $u^4 A = A$ implies

$$u^4 = 1.$$

If $u = \pm 1$, then $u^6 = 1$, and the second equation in (11) becomes

$$s(s^2 + A) = 0.$$

Since $\mathbb{P}^1$ is integral and $s \neq 0$, multiplication by s is injective on the section ring. Hence

$$A = -s^2.$$

If $u^2 = -1$, then $u^6 = -1$, and therefore

$$u^6 - 1 = -2 = 1$$

in characteristic three. Thus

$$C = s^3 + As.$$

The group law gives

$$(1, s)^3 = (1, 0), \quad (-1, s)^3 = (-1, 0),$$

for $s \neq 0$. Hence $(1, s)$ has order three and $(-1, s)$ has order six. If $u^2 = -1$, then

$$(u, s)^2 = (-1, 0),$$

so (u, s) has order four. Finally, -1 is a square in $\mathbb{F}_{3^r}$ exactly when r is even. $\blacksquare$

3.1. The ambient marked count. For a quotient $[U/G]$, its inertia records pairs (z, g) with $z \in U$ and $g \in G$ satisfying $g \cdot z = z$. Thus g is a marked automorphism of z . We call the count below *ambient* because it is taken in coefficient degree d , before relative minimality and intrinsic height are imposed.

Let

$$Z_d^{(3)} = \{(z, g) \in \mathcal{U}_d^{(3),0} \times G_d^{(3)} : g \cdot z = z\}$$

be the full fixed-pair incidence scheme. The identity and inversion define disjoint sections

$$\mathbf{e}(A, C) = ((A, C), (1, 0))$$

and

$$\mathbf{i}(A, C) = ((A, C), (-1, 0))$$

of $Z_d^{(3)} \rightarrow \mathcal{U}_d^{(3),0}$. Define the extra fixed-pair incidence scheme by

$$Z_d^{(3),\text{ex}} = Z_d^{(3)} \setminus (\mathbf{e}(\mathcal{U}_d^{(3),0}) \sqcup \mathbf{i}(\mathcal{U}_d^{(3),0})).$$

The group $G_d^{(3)}$ acts on this incidence by change of coordinates and conjugation:

$$h \cdot (z, g) = (h \cdot z, hgh^{-1}).$$

Put

$$Y_d^{(3)}(q) = \#_q[Z_d^{(3),\text{ex}}/G_d^{(3)}](\mathbb{F}_q).$$

This is the weighted count of coefficient-degree- d equations marked by an automorphism other than 1 and $[-1]$, before relative minimality is imposed.

Proposition 3.3. *For every $d \geq 0$,*

$$(12) \quad Y_d^{(3)}(q) = 2P_d^{(3)}(q) + 2\epsilon_r Q_d^{(3)}(q),$$

where ϵ_r is the parity indicator defined in (1), and

$$(13) \quad P_d^{(3)}(q) = q^{4d} \frac{q^{2d+1} - 1}{q - 1}, \quad Q_d^{(3)}(q) = \frac{q^{4d+1} - 1}{q - 1}.$$

Proof. Since

$$H^1(\mathbb{F}_q, G_d^{(3)}) = \{1\},$$

the groupoid

$$[Z_d^{(3),\text{ex}}/G_d^{(3)}](\mathbb{F}_q)$$

is equivalent to the ordinary action groupoid of $G_d^{(3)}(\mathbb{F}_q)$ on $Z_d^{(3),\text{ex}}(\mathbb{F}_q)$. The orbit-stabilizer formula therefore gives

$$Y_d^{(3)}(q) = \frac{|Z_d^{(3),\text{ex}}(\mathbb{F}_q)|}{|G_d^{(3)}(\mathbb{F}_q)|}.$$

Fix $u \in \{1, -1\}$. By Lemma 3.2, the $\mathbb{F}_q$ -points of the fixed-pair locus with this value of u are determined by

$$0 \neq s \in V_2(d), \quad C \in V_6(d),$$

with $A = -s^2$. There are

$$(q^{2d+1} - 1)q^{6d+1}$$

such fixed pairs. Dividing by

$$|G_d^{(3)}(\mathbb{F}_q)| = (q-1)q^{2d+1}$$

gives $P_d^{(3)}(q)$ for each of the two values of u .

Suppose that r is even and fix

$$u \in \mathbb{F}_q \quad \text{with} \quad u^2 = -1.$$

The $\mathbb{F}_q$ -points of the corresponding order-four fixed-pair locus are determined by

$$0 \neq A \in V_4(d), \quad s \in V_2(d),$$

with $C = s^3 + As$. There are

$$(q^{4d+1} - 1)q^{2d+1}$$

such fixed pairs. Dividing by $|G_d^{(3)}(\mathbb{F}_q)|$ gives $Q_d^{(3)}(q)$ for each of the two roots. When r is odd, the two geometric order-four fixed-pair loci have no $\mathbb{F}_q$ -points and contribute zero.

The projection

$$G_d^{(3)} \longrightarrow \mathbb{G}_m$$

is invariant under conjugation. Consequently, each fixed-pair locus with prescribed value of u is $G_d^{(3)}$ -stable, and its contribution may be computed separately.

The stabilizer of an $\mathbb{F}_q$ -valued fixed pair (z, g) is

$$C_{\text{Stab}_{G_d^{(3)}(\mathbb{F}_q)}(z)}(g).$$

By Proposition 3.1, this is the centralizer of the marked automorphism in the automorphism group of the corresponding resolved model. Hence orbit-stabilizer gives the required marked-inertia weights. Adding the contributions of the two loci indexed by $u = \pm 1$ and, when r is even, the two loci indexed by $u^2 = -1$ proves (12). ■

4. EXTRA AUTOMORPHISMS IN CHARACTERISTIC 2

Assume throughout this section that $q = 2^r$ with $r \geq 1$. Retain the notation

$$V_i(d) = H^0(\mathbb{P}^1, \mathcal{O}(id)), \quad i, d \geq 0.$$

4.1. Normalized equations and the residual coordinate group. A generically smooth $j = 0$ equation of coefficient degree d has the normalized form

$$(14) \quad E_{A,B,C} : y^2 + Ay = x^3 + Bx + C,$$

where

$$0 \neq A \in V_3(d), \quad B \in V_4(d), \quad C \in V_6(d).$$

Its discriminant is A^4 , so $A \neq 0$ is precisely the condition of generic smoothness. Put

$$\mathcal{U}_d^{(2),0} = (V_3(d) \setminus \{0\}) \times V_4(d) \times V_6(d).$$

The *residual coordinate group*, namely the group of global Weierstrass coordinate changes remaining after normalization and preserving (14), has underlying scheme

$$G_d^{(2)} = \mathbb{G}_m \times V_1(d) \times V_3(d),$$

but its multiplication is not the direct-product group law. For $g = (u, s, t)$, define the coordinate change

$$(15) \quad \Phi_g(x', y') = (u^2x' + s^2, u^3y' + u^2sx' + t).$$

The multiplication determined by

$$\Phi_{g_1 g_2} = \Phi_{g_1} \circ \Phi_{g_2}$$

is

$$(16) \quad \begin{aligned} & (u_1, s_1, t_1)(u_2, s_2, t_2) \\ &= (u_1 u_2, u_1 s_2 + s_1, u_1^3 t_2 + u_1^2 s_1 s_2^2 + t_1), \end{aligned}$$

and

$$(17) \quad (u, s, t)^{-1} = (u^{-1}, u^{-1}s, u^{-3}(t + s^3)).$$

Pullback by (15) gives the right action

$$(18) \quad \begin{aligned} (A, B, C) \triangleleft (u, s, t) &= (u^{-3}A, u^{-4}(B + As + s^4), \\ &u^{-6}(C + Bs^2 + s^6 + At + t^2)). \end{aligned}$$

It is a right action in the sense that

$$((A, B, C) \triangleleft g_1) \triangleleft g_2 = (A, B, C) \triangleleft (g_1 g_2).$$

We write $[\mathcal{U}_d^{(2),0}/G_d^{(2)}]$ for the quotient by this right action; equivalently, it is the quotient for the associated left action

$$g \cdot z = z \triangleleft g^{-1}.$$

For fixed pairs we use the equivalent right-action convention below. Under passage to the associated left action, a right-fixed element g corresponds to the marked element g^{-1} . This replacement preserves the order and the centralizer of the marked automorphism.

Proposition 4.1. *Let $\mathcal{W}_d^{(2),0}(q)$ be the groupoid of generically smooth $j = 0$ Weierstrass equations over the fixed base $\mathbb{P}_{\mathbb{F}_q}^1$ of coefficient degree d , with morphisms given by global Weierstrass coordinate changes over the identity of $\mathbb{P}_{\mathbb{F}_q}^1$. Normalization induces an equivalence*

$$\mathcal{W}_d^{(2),0}(q) \simeq [\mathcal{U}_d^{(2),0}/G_d^{(2)}](\mathbb{F}_q).$$

Let

$$\bar{X}_{A,B,C} \longrightarrow \mathbb{P}^1$$

be the corresponding coefficient-degree- d Weierstrass surface, and let

$$\tilde{X}_{A,B,C} \longrightarrow \bar{X}_{A,B,C}$$

be its minimal desingularization, with lifted zero section $\tilde{\sigma}_{A,B,C}$. This model need not be relatively-minimal. Then

$$\text{Stab}_{G_d^{(2)}(\mathbb{F}_q)}(A, B, C) \simeq \text{Aut}_{\mathbb{P}^1}(\tilde{X}_{A,B,C}, \tilde{\sigma}_{A,B,C}) \simeq \text{Aut}_K(E_{A,B,C}, O).$$

Moreover, $G_d^{(2)}$ is smooth and connected,

$$(19) \quad |G_d^{(2)}(\mathbb{F}_q)| = (q-1)q^{4d+2}, \quad H^1(\mathbb{F}_q, G_d^{(2)}) = \{1\}.$$

Consequently, there is an equivalence

$$[\mathcal{U}_d^{(2),0}/G_d^{(2)}](\mathbb{F}_q) \simeq \mathcal{U}_d^{(2),0}(\mathbb{F}_q) // G_d^{(2)}(\mathbb{F}_q).$$

Proof. For a generalized Weierstrass equation in characteristic two, the conditions $j = 0$ and generic smoothness force $a_1 = 0$. de Jong's normalization eliminates the x^2 coefficient [dJ02, Section 4.13(c2)], giving (14).

Every isomorphism of generalized Weierstrass equations is induced by an admissible coordinate change [dJ02, Sections 4.8–4.9]; see also [Sil09, Chapter III, Section 1]. Between two normalized equations it has the form

$$x = u^2x' + \rho, \quad y = u^3y' + u^2\eta x' + t,$$

where

$$u \in \mathbb{F}_q^\times, \quad \rho \in V_2(d), \quad \eta \in V_1(d), \quad t \in V_3(d).$$

Comparison of the x'^2 coefficients gives

$$\rho = \eta^2.$$

Writing $s = \eta$ therefore gives (15). Direct composition and substitution give (16)–(18). This identifies the normalized-equation groupoid with the ordinary action groupoid. Lang's theorem below identifies the latter with the $\mathbb{F}_q$ -points of the quotient stack.

Let $X^{\text{rel}} \rightarrow \mathbb{P}^1$ be the relatively-minimal regular model of the generic fiber. Restriction to the generic fiber identifies

$$\text{Aut}_{\mathbb{P}^1}(X^{\text{rel}}, \sigma) \simeq \text{Aut}_K(E_{A,B,C}, O)$$

[dJ02, Section 4.2]. de Jong's construction obtains $\tilde{X}_{A,B,C}$ from X^{rel} by the prescribed sequence of blowups at closed points on the successive strict transforms of the zero section. Every automorphism fixes the zero section pointwise and hence preserves each blowup center scheme-theoretically. The universal property of the blowup therefore gives a unique lift through every stage of the construction.

Conversely, an automorphism of $(\tilde{X}_{A,B,C}, \tilde{\sigma}_{A,B,C})$ restricts to an automorphism of the generic fiber. The lift of that restriction agrees with the original automorphism on the generic fiber. Since $\tilde{X}_{A,B,C}$ is integral and the target is separated, the two morphisms agree everywhere. de Jong's coordinate-choice construction then identifies this automorphism group with the stabilizer of (A, B, C) [dJ02, Sections 4.8–4.11].

The displayed description of the underlying scheme shows that $G_d^{(2)}$ is smooth and connected. Since

$$\dim_{\mathbb{F}_q} V_i(d) = id + 1,$$

one has

$$|G_d^{(2)}(\mathbb{F}_q)| = (q-1)q^{(d+1)+(3d+1)} = (q-1)q^{4d+2}.$$

Lang's theorem gives

$$H^1(\mathbb{F}_q, G_d^{(2)}) = \{1\},$$

which proves the final equivalence. ■

For comparison, the weighted cardinality of the entire generically smooth normalized $j = 0$ locus of coefficient degree d is

$$(20) \quad \frac{(q^{3d+1} - 1)q^{4d+1}q^{6d+1}}{(q-1)q^{4d+2}} = q^{6d} \frac{q^{3d+1} - 1}{q-1}.$$

It has leading growth q^{9d} . By contrast, the ambient marked count computed below has leading growth q^{5d} when r is odd and q^{6d} when r is even.

4.2. Classification of fixed pairs. A *fixed pair* consists of a normalized coefficient tuple $z = (A, B, C) \in \mathcal{U}_d^{(2),0}$ together with an element $g \in G_d^{(2)}$ satisfying

$$z \triangleleft g = z,$$

where $\triangleleft$ denotes the right coefficient action defined in (18).

For $g = (u, s, t) \in G_d^{(2)}(\mathbb{F}_q)$, put

$$v = u^2s.$$

Lemma 4.2. *The residual coordinate change $g = (u, s, t)$ stabilizes $E_{A,B,C}$ if and only if*

$$(21) \quad u^3 = 1$$

and, after setting $v = u^2s$ so that $s = uv$, it acts as

$$(22) \quad \phi_{u,v,t}(x, y) = (u^2x + u^2v^2, y + vx + t)$$

and satisfies

$$(23) \quad v^4 + Av + (1 + u^2)B = 0,$$

$$(24) \quad t^2 + At = v^6 + u^2Bv^2.$$

In these coordinates, inversion is the nontrivial involution

$$[-1](x, y) = (x, y + A),$$

represented by $(u, v, t) = (1, 0, A)$. Apart from the identity $(1, 0, 0)$ and this inversion, the $\mathbb{F}_q$ -defined solutions are the following.

(I) Suppose that $u = 1$ and $v \neq 0$. There is a unique $w \in V_2(d)$ such that

$$(25) \quad A = v^3, \quad t = vw, \quad B = w^2 + v^2w + v^4,$$

while C is arbitrary. The marked automorphism has order four and its square is inversion.

(II) Suppose that

$$u \in \mu_3(\mathbb{F}_q) \setminus \{1\},$$

where

$$\mu_3(\mathbb{F}_q) = \{a \in \mathbb{F}_q^\times : a^3 = 1\}$$

is the group of cube roots of unity in $\mathbb{F}_q$. The two nontrivial cube roots of unity belong to $\mathbb{F}_q$ if and only if r is even. For each such u , the sections

$$A \in V_3(d) \setminus \{0\}, \quad v \in V_1(d)$$

are arbitrary,

$$(26) \quad B = u^2(v^4 + Av),$$

and

$$(27) \quad t = uv^3 + \varepsilon A, \quad \varepsilon \in \{0, 1\}.$$

For $\varepsilon = 0$, the marked automorphism has order three. For $\varepsilon = 1$, it has order six and its cube is inversion. Again C is arbitrary.

Proof. Equating (18) with (A, B, C) gives

$$(u^3 - 1)A = 0,$$

$$(u^4 - 1)B + As + s^4 = 0,$$

and

$$(u^6 - 1)C + Bs^2 + s^6 + At + t^2 = 0.$$

Since $A \neq 0$, the first equation gives $u^3 = 1$. Substitution of $v = u^2s$ then gives (23) and (24).

Suppose first that $u = 1$. If $v = 0$, then

$$t(t + A) = 0.$$

Since $\mathbb{P}^1$ is integral and $A \neq 0$, this gives $t = 0$ or $t = A$, corresponding respectively to the identity and inversion.

Now suppose that $v \neq 0$. Equation (23) becomes

$$v(v^3 + A) = 0.$$

Since $\mathbb{P}^1$ is integral and $v \neq 0$, multiplication by v is injective on global sections, and hence

$$A = v^3.$$

We claim that v divides t . If v is nowhere vanishing, then $d = 0$ and the assertion is immediate. Otherwise, let x be any zero of v of order $a > 0$. If

$$\text{ord}_x(t) = b < a,$$

then

$$\text{ord}_x(t^2 + At) = 2b < 2a,$$

whereas

$$\text{ord}_x(v^6 + Bv^2) \geq 2a,$$

a contradiction. Thus

$$\text{ord}_x(t) \geq \text{ord}_x(v)$$

at every zero x of v . Hence t/v has no poles and defines a unique global section

$$w = \frac{t}{v} \in V_2(d).$$

Thus

$$t = vw.$$

Dividing (24) by v^2 gives (25).

Now suppose that $u \neq 1$. Then

$$u^2 + u + 1 = 0.$$

Equation (23) gives (26). After this substitution, (24) becomes

$$(t + uv^3)(t + uv^3 + A) = 0.$$

Since $\mathbb{P}^1$ is integral, one of these two factors vanishes identically, giving (27). The two nontrivial cube roots of unity belong to $\mathbb{F}_{2^r}$ if and only if r is even.

Finally, direct composition gives

$$\phi_{1,v,t}^2(x, y) = (x, y + v^3) = (x, y + A)$$

in case (I), so the square is inversion. In case (II), one obtains

$$\phi_{u,v,t}^3(x, y) = (x, y + t + uv^3).$$

This is the identity when $\varepsilon = 0$ and inversion when $\varepsilon = 1$, proving the asserted orders. ■

Remark 4.3. Lemma 4.2 classifies $\mathbb{F}_q$ -valued fixed pairs. It does not identify the full fixed-pair incidence scheme with the reduced parameter spaces in (25)–(27).

Indeed, the type-(I) formulas define the parametrization

$$(v, w, C) \longmapsto ((A, B, C), (u, v, t)) = ((v^3, w^2 + v^2w + v^4, C), (1, v, vw)), \quad v \neq 0.$$

We call its image the *type-(I) parameterizing graph*. The following calculation exhibits excess tangent directions along this graph when $d > 0$.

At a point

$$u = 1, \quad A = v^3, \quad t = vw,$$

the linearized equations, after separating the unrestricted δC -direction, give

$$\delta u = 0, \quad \delta A = v^2 \delta v, \quad \delta B = v \delta t + vw \delta v.$$

The tangent space to the parameterizing graph additionally requires

$$\delta t = w \delta v + v \delta w.$$

Thus the excess tangent quotient is represented by

$$[\delta t - w \delta v] \in V_3(d)/vV_2(d).$$

Since multiplication by the nonzero section v is injective, this quotient has dimension

$$(3d + 1) - (2d + 1) = d.$$

Because $\mathbb{F}_q$ is reduced, a finite-type $\mathbb{F}_q$ -scheme and its reduction have the same set of $\mathbb{F}_q$ -points. Thus these excess tangent directions do not change the set of $\mathbb{F}_q$ -valued fixed pairs or the corresponding action-groupoid cardinality. This phenomenon concerns the incidence in which both the equation and the marked automorphism vary; it does not contradict Remark 2.2, which concerns the automorphism group scheme of a fixed generic fiber.

4.3. The ambient marked count. Let

$$Z_d^{(2)} = \{(z, g) \in \mathcal{U}_d^{(2),0} \times G_d^{(2)} : z \triangleleft g = z\}$$

be the full fixed-pair incidence scheme. The identity and inversion define disjoint sections

$$\mathbf{e}(A, B, C) = ((A, B, C), (1, 0, 0))$$

and

$$\iota(A, B, C) = ((A, B, C), (1, 0, A))$$

of $Z_d^{(2)} \rightarrow \mathcal{U}_d^{(2),0}$. Define the extra fixed-pair incidence scheme by

$$Z_d^{(2),\text{ex}} = Z_d^{(2)} \setminus (\mathbf{e}(\mathcal{U}_d^{(2),0}) \sqcup \iota(\mathcal{U}_d^{(2),0})).$$

The group $G_d^{(2)}$ acts on this incidence by change of coordinates and conjugation:

$$(z, g) \triangleleft h = (z \triangleleft h, h^{-1}gh).$$

Put

$$Y_d^{(2)}(q) = \#_q[Z_d^{(2),\text{ex}}/G_d^{(2)}](\mathbb{F}_q).$$

This is the weighted count of coefficient-degree- d equations marked by an automorphism other than 1 and $[-1]$, before relative minimality is imposed.

Proposition 4.4. *For every $d \geq 0$,*

$$(28) \quad Y_d^{(2)}(q) = P_d^{(2)}(q) + 4\epsilon_r Q_d^{(2)}(q),$$

where ϵ_r is the parity indicator defined in (1), and

$$(29) \quad P_d^{(2)}(q) = q^{4d} \frac{q^{d+1} - 1}{q - 1}, \quad Q_d^{(2)}(q) = q^{3d} \frac{q^{3d+1} - 1}{q - 1}.$$

Proof. By Proposition 4.1,

$$H^1(\mathbb{F}_q, G_d^{(2)}) = \{1\}.$$

Hence

$$[Z_d^{(2),\text{ex}}/G_d^{(2)}](\mathbb{F}_q) \simeq Z_d^{(2),\text{ex}}(\mathbb{F}_q) // G_d^{(2)}(\mathbb{F}_q).$$

The orbit-stabilizer formula and (19) therefore give

$$Y_d^{(2)}(q) = \frac{|Z_d^{(2),\text{ex}}(\mathbb{F}_q)|}{|G_d^{(2)}(\mathbb{F}_q)|}.$$

The excess tangent directions described in Remark 4.3 do not affect the numerator.

For type (I), an $\mathbb{F}_q$ -valued fixed pair is determined by

$$0 \neq v \in V_1(d), \quad w \in V_2(d), \quad C \in V_6(d).$$

There are

$$(q^{d+1} - 1)q^{2d+1}q^{6d+1}$$

such pairs. Dividing by

$$|G_d^{(2)}(\mathbb{F}_q)| = (q-1)q^{4d+2}$$

gives $P_d^{(2)}(q)$.

Suppose that r is even. For each choice of

$$u \in \mu_3(\mathbb{F}_q) \setminus \{1\}, \quad \varepsilon \in \{0, 1\},$$

a type-(II) fixed pair is determined by

$$0 \neq A \in V_3(d), \quad v \in V_1(d), \quad C \in V_6(d),$$

with B and t forced by (26) and (27). There are

$$(q^{3d+1} - 1)q^{d+1}q^{6d+1}$$

such pairs for each fixed (u, ε) . Division by $|G_d^{(2)}(\mathbb{F}_q)|$ gives $Q_d^{(2)}(q)$. There are two choices of u and two choices of ε , giving four fixed-pair loci and hence the factor $4\epsilon_r$. When r is odd, one has

$$\mu_3(\mathbb{F}_q) = \{1\},$$

so there are no $\mathbb{F}_q$ -valued type-(II) fixed pairs.

It remains to verify the weights. For the right action, the stabilizer of an $\mathbb{F}_q$ -valued fixed pair (z, g) is

$$C_{\text{Stab}_{G_d^{(2)}(\mathbb{F}_q)}(z)}(g).$$

By Proposition 4.1, this is the centralizer of the marked automorphism in the automorphism group of the generic fiber corresponding to z . Orbit-stabilizer therefore weights each isomorphism class of marked coefficient-degree- d models by the reciprocal of the corresponding centralizer. Thus the quotient has precisely the required marked-inertia weights. $\blacksquare$

5. MARKED AUTOMORPHISMS AND MINIMALITY DEFECT

For $p \in \{2, 3\}$ and $r \geq 1$, let $q = p^r$. We use a superscript (p) on the normalized fixed-pair spaces, residual coordinate groups, and ambient marked counts constructed in the preceding two sections.

The construction below is the marked, characteristic-two-and-three analogue of the minimality-defect stratification of weighted linear series developed by Bejleri–Park–Satriano. In their setting, if $\lambda_0, \dots, \lambda_N$ are the weights and

$$\kappa = \text{lcm}(\lambda_0, \dots, \lambda_N),$$

the minimality defect is the size of the quotient obtained by dividing the normalized base profile by κ [BPS22, Definition 4.31]. The corresponding normalized base-locus divisor admits a canonical quotient-remainder decomposition

$$D_{\text{base}} = \kappa D_{\text{def}} + D_{\text{rem}}, \quad \deg D_{\text{def}} = e$$

[BPS22, Lemma 4.32]. Over $\mathbb{P}^1$, after a locally closed stratification, the defect- e stratum is obtained from a minimal object of height smaller by e , together with an effective divisor in $\text{Sym}^e(\mathbb{P}^1)$ [BPS22, Proposition 6.1 and Corollary 6.2]. The corresponding motivic ordinary and inertia convolutions are recorded in [BPS22, Lemma 8.6, equations (6)–(7)].

We follow the same sequence for normalized $j = 0$ models in characteristics 2 and 3: recover the defect divisor, establish the marked defect decomposition, and invert the resulting divisor convolution. The representative-level coefficient factorization used in [BPS22, Section 6.1] cannot be imposed invariantly on the normalized coefficients occurring here, because of the residual translations in these characteristics. We therefore formulate the defect intrinsically on the resolved elliptic surface and verify that lifting and contraction preserve the centralizer of the marked automorphism.

A *minimal desingularization* resolves the singularities of a chosen Weierstrass surface, but it need not be relatively-minimal as an elliptic fibration. A curve on an elliptic surface is *vertical* if it is contained in a fiber, and a (-1) -curve is a smooth rational curve of self-intersection -1 . Relative minimalization contracts the vertical (-1) -curves arising from the defect. We will show that a resolved coefficient-degree- d model is determined by its relatively-minimal model of exact Faltings height n , together with an effective divisor D of degree

$$\deg D = d - n.$$

Let $\mathcal{A}_d^{(p),\text{ex}}(q)$ be the essentially finite groupoid of resolved coefficient-degree- d models with a marked extra automorphism. Thus an object is a triple

$$(\tilde{X}, \tilde{\sigma}, \alpha)$$

such that $(\tilde{X}, \tilde{\sigma})$ is isomorphic over $\mathbb{P}_{\mathbb{F}_q}^1$ to the minimal desingularization of a generically smooth normalized $j = 0$ Weierstrass equation of coefficient degree d , and

$$\alpha \in \text{Aut}_{\mathbb{P}^1}(\tilde{X}, \tilde{\sigma}) \setminus \{1, [-1]\}.$$

Here $[-1]$ denotes the unique extension to the resolved pair of inversion on the generic fiber. An arrow

$$\varphi : (\tilde{X}, \tilde{\sigma}, \alpha) \longrightarrow (\tilde{X}', \tilde{\sigma}', \alpha')$$

is an isomorphism over the identity of $\mathbb{P}_{\mathbb{F}_q}^1$ satisfying

$$\varphi \circ \tilde{\sigma} = \tilde{\sigma}', \quad \varphi \alpha \varphi^{-1} = \alpha'.$$

The conormal bundle of the zero section,

$$\tilde{\sigma}^* \mathcal{O}_{\tilde{X}}(-\tilde{\sigma}),$$

has degree d .

Let $\mathcal{M}_n^{\text{ex}}(q)$ be the essentially finite groupoid whose objects are triples

$$(X, \sigma, \alpha),$$

where (X, σ) is a relatively-minimal proper regular elliptic surface over $\mathbb{P}_{\mathbb{F}_q}^1$ of exact Faltings height n and

$$\alpha \in \text{Aut}_{\mathbb{P}^1}(X, \sigma) \setminus \{1, [-1]\}.$$

An arrow

$$\varphi : (X, \sigma, \alpha) \longrightarrow (X', \sigma', \alpha')$$

is an isomorphism over the identity of $\mathbb{P}_{\mathbb{F}_q}^1$ satisfying

$$\varphi \circ \sigma = \sigma', \quad \varphi \alpha \varphi^{-1} = \alpha'.$$

Thus, in either groupoid, the automorphism group of a marked object is the centralizer of its marked automorphism in the automorphism group of the underlying pointed surface.

Restriction to the generic fiber identifies the groupoid of relatively minimal elliptic surfaces over the fixed base with the groupoid of elliptic curves over $K = \mathbb{F}_q(t)$ and preserves automorphism groups [dJ02, Section 4.2]. In particular,

$$(30) \quad \text{Aut}_{\mathbb{P}^1}(X, \sigma) \simeq \text{Aut}_K(E, O),$$

and therefore

$$\#_q \mathcal{M}_n^{\text{ex}}(q) = X_n(q).$$

Lemma 5.1. *For every $p \in \{2, 3\}$ and $d \geq 0$, there are equivalences*

$$(31) \quad [Z_d^{(p), \text{ex}} / G_d^{(p)}](\mathbb{F}_q) \simeq Z_d^{(p), \text{ex}}(\mathbb{F}_q) // G_d^{(p)}(\mathbb{F}_q) \simeq \mathcal{A}_d^{(p), \text{ex}}(q).$$

Consequently,

$$Y_d^{(p)}(q) = \#_q \mathcal{A}_d^{(p), \text{ex}}(q).$$

Proof. The explicit descriptions of the residual coordinate groups show that $G_d^{(p)}$ is a smooth connected linear algebraic group over $\mathbb{F}_q$. By Lang's theorem, every $G_d^{(p)}$ -torsor over $\mathbb{F}_q$ is trivial. Consequently, every $\mathbb{F}_q$ -object of the quotient stack has a representative over $\mathbb{F}_q$. Hence the natural functor

$$Z_d^{(p), \text{ex}}(\mathbb{F}_q) // G_d^{(p)}(\mathbb{F}_q) \longrightarrow [Z_d^{(p), \text{ex}} / G_d^{(p)}](\mathbb{F}_q)$$

is an equivalence.

Propositions 3.1 and 4.1 identify the resulting action groupoid with the groupoid of normalized coefficient-degree- d equations marked by an extra automorphism. Minimal desingularization therefore defines a functor to $\mathcal{A}_d^{(p), \text{ex}}(q)$.

A coordinate change between Weierstrass equations lifts uniquely to their minimal desingularizations. Conversely, the *canonical Weierstrass model* of a resolved pair is obtained by contracting, in every fiber, the irreducible components disjoint from the zero section. This contraction is functorial in the resolved pair and its zero section. If the resolved pair is the minimal desingularization of a Weierstrass model, the contraction recovers that model. Hence an isomorphism of resolved pairs descends to an isomorphism of their canonical Weierstrass models [dJ02, Sections 4.7–4.9]. In particular, the marked automorphism of any object of $\mathcal{A}_d^{(p), \text{ex}}(q)$ descends to its canonical Weierstrass model. After choosing normalized presentations of the canonical models, Propositions 3.1 and 4.1 identify the descended automorphisms and isomorphisms with elements of $G_d^{(p)}(\mathbb{F}_q)$. Hence the functor is essentially surjective and fully faithful.

These constructions preserve the marked automorphism and carry conjugation to conjugation. In particular, they identify the centralizers of corresponding marked elements, so they preserve weighted cardinality. ■

Recall that

$$c_e(q) = |\text{Sym}^e(\mathbb{P}^1)(\mathbb{F}_q)| = \frac{q^{e+1} - 1}{q - 1}.$$

An $\mathbb{F}_q$ -point of $\text{Sym}^e(\mathbb{P}^1)$ is equivalently a Frobenius-stable effective divisor

$$D = \sum_{x \in |\mathbb{P}^1|} e_x x$$

of degree

$$\deg D = \sum_{x \in |\mathbb{P}^1|} e_x \deg(x) = e.$$

This is the effective divisor in de Jong's blowup description. Each blowup above a closed point x increases the degree of the section conormal bundle by $\deg(x)$. By analogy with [BPS22, Definition 4.31], we call the integer

$$e = d - n = \deg D$$

the *minimality defect*. More precisely, our divisor D plays the role of the effective quotient divisor in [BPS22, Lemma 4.32]. It is intrinsic, rather than being determined by visible divisibility of the coefficients in a chosen normalized equation.

The map $\psi_{n,e}$ of [BPS22, Section 6.1 and Proposition 6.1] is written on representatives as

$$(f_0, \dots, f_N) = (g_0 h^{\lambda_0}, \dots, g_N h^{\lambda_N}),$$

where h is a degree- e form. The following lemma gives the intrinsic resolved-surface analogue of this factorization.

Lemma 5.2. *Let*

$$f : (X, \sigma) \longrightarrow \mathbb{P}_{\mathbb{F}_q}^1$$

be a relatively-minimal proper regular elliptic surface of exact Faltings height n . Write

$$\mathcal{L}_f = (R^1 f_* \mathcal{O}_X)^\vee$$

for its fundamental line bundle, and let

$$D = \sum_{x \in |\mathbb{P}^1|} e_x x$$

be an effective divisor. Put

$$d = n + \deg D.$$

For each closed point x , blow up $\sigma(x)$ and then the closed point above x on each successive strict transform of the section, a total of e_x times. Denote the resulting pair by

$$(\tilde{X}_D, \tilde{\sigma}_D).$$

Then

$$(32) \quad \tilde{\sigma}_D^* \mathcal{O}_{\tilde{X}_D}(-\tilde{\sigma}_D) \simeq \mathcal{L}_f \otimes \mathcal{O}_{\mathbb{P}^1}(D),$$

so this line bundle has degree d . The canonical Weierstrass model associated to $(\tilde{X}_D, \tilde{\sigma}_D)$ is of coefficient degree d , and $\tilde{X}_D$ is its minimal desingularization.

The construction is functorial under isomorphisms over the identity of $\mathbb{P}_{\mathbb{F}_q}^1$. In particular, lifting induces an isomorphism

$$\text{Aut}_{\mathbb{P}^1}(X, \sigma) \simeq \text{Aut}_{\mathbb{P}^1}(\tilde{X}_D, \tilde{\sigma}_D)$$

that carries 1 and $[-1]$ to their corresponding lifts, respects conjugation, and hence identifies the centralizers of corresponding marked automorphisms.

Suppose that $p = 2$ or 3 and that the generic fiber has $j = 0$. The corresponding normalization preserves the isomorphism class of the resolved pair and the centralizer of a marked automorphism.

Proof. The section self-intersection formula gives

$$\sigma^* \mathcal{O}_X(-\sigma) \simeq \mathcal{L}_f, \quad \deg \mathcal{L}_f = n.$$

Blowing up the section above a closed point x tensors the conormal bundle of its strict transform by $\mathcal{O}_{\mathbb{P}^1}(x)$. Iterating over D proves (32), whose degree is

$$n + \deg D = d.$$

We verify the minimal-resolution criterion of [dJ02, Section 4.7]. After base change to $\overline{\mathbb{F}}_q$, fix a geometric point $\bar{x}$ above a closed point x with $e_x > 0$. The zero section lies in the smooth locus of f , so $\sigma(\bar{x})$ is a smooth point of the fiber. Write

$$E_{\bar{x},1}, \dots, E_{\bar{x},e_x}$$

for the successive exceptional curves. For $1 \leq i < e_x$, the curve $E_{\bar{x},i}$ is blown up exactly once more, at its intersection with the successive strict transform of the section. Its final strict transform is therefore a vertical (-2) -curve disjoint from the section. The terminal curve $E_{\bar{x},e_x}$ is a vertical (-1) -curve meeting $\tilde{\sigma}_D$.

No strict transform of a vertical curve on $X_{\overline{\mathbb{F}}_q}$ produces an additional (-1) -curve. Indeed, every smooth rational component of a reducible fiber has self-intersection at most -2 by relative minimality, and blowing up can only decrease its self-intersection. If the fiber through $\bar{x}$ is irreducible, its strict transform after the first blowup has self-intersection -1 , but it remains a curve of arithmetic genus one and hence is not a (-1) -curve. All subsequent centers above $\bar{x}$ lie on exceptional curves. Thus the terminal exceptional curves are precisely the vertical (-1) -curves of $\tilde{X}_D$, and all of them meet the section.

The criterion of [dJ02, Section 4.7] now shows that $\tilde{X}_D$ is the minimal desingularization of its canonical Weierstrass contraction. de Jong's coordinate construction [dJ02, Sections 4.5–4.6] writes this Weierstrass model with coefficients

$$a_i \in H^0(\mathbb{P}^1, (\tilde{\sigma}_D^* \mathcal{O}_{\tilde{X}_D}(-\tilde{\sigma}_D))^{\otimes i}).$$

Every line bundle of degree d on $\mathbb{P}_{\overline{\mathbb{F}}_q}^1$ is isomorphic to $\mathcal{O}(d)$, so the model is of coefficient degree d .

Let

$$\varphi: (X, \sigma) \longrightarrow (X', \sigma')$$

be an isomorphism over the identity of $\mathbb{P}_{\overline{\mathbb{F}}_q}^1$, and use the same divisor D on both sides. Since $\varphi \circ \sigma = \sigma'$, it carries every closed and infinitely near blowup center scheme-theoretically to the corresponding center. The universal property of the blowup gives a unique lift

$$\tilde{\varphi}_D: (\tilde{X}_D, \tilde{\sigma}_D) \longrightarrow (\tilde{X}'_D, \tilde{\sigma}'_D).$$

This lifting construction is compatible with identities and composition.

Conversely, after base change to $\overline{\mathbb{F}}_q$, the terminal exceptional curves are characterized intrinsically as the vertical (-1) -curves of the blown-up surface. They form Frobenius-stable disjoint collections indexed by the closed points x for which

$e_x > 0$. The simultaneous contraction of each such collection is unique, hence Frobenius equivariant, and therefore descends to $\mathbb{F}_q$.

Any isomorphism of the blown-up pairs preserves these collections and descends uniquely through their contractions. After the terminal collection above each x is contracted, the preceding exceptional curves for those x with $e_x \geq 2$ become the new terminal geometric (-1) -curves. Iterating recovers (X, σ) and all multiplicities e_x . Thus lifting and contraction are mutually inverse on isomorphisms.

The resulting isomorphism of automorphism groups respects composition and conjugation and carries 1 and $[-1]$ to the corresponding elements; it therefore identifies the stated centralizers.

Finally, suppose that $p \in \{2, 3\}$ and that the generic fiber has $j = 0$. The normalizations in Propositions 3.1 and 4.1 are effected by global admissible Weierstrass coordinate changes. Such a change induces an isomorphism of the associated Weierstrass models and hence, by uniqueness of minimal desingularizations, an isomorphism of their resolved pairs. Transport through this isomorphism carries a marked automorphism to the corresponding marked automorphism and preserves its centralizer. ■

The preceding lemma now yields the marked defect decomposition.

Proposition 5.3. *For every $p \in \{2, 3\}$ and $d \geq 0$, relative minimalization, together with recovery of the defect divisor, induces an equivalence of essentially finite groupoids*

$$(33) \quad \mathcal{A}_d^{(p), \text{ex}}(q) \simeq \bigsqcup_{n=0}^d \bigsqcup_{D \in \text{Sym}^{d-n}(\mathbb{P}^1)(\mathbb{F}_q)} \mathcal{M}_n^{\text{ex}}(q),$$

where each symmetric-power set is regarded as a discrete groupoid, meaning that it has only identity arrows. This equivalence identifies the full automorphism groups and the centralizers of corresponding marked automorphisms. Consequently,

$$(34) \quad Y_d^{(p)}(q) = \sum_{n=0}^d c_{d-n}(q) X_n(q) = \sum_{e=0}^d c_e(q) X_{d-e}(q).$$

Proof. Let

$$(\tilde{X}, \tilde{\sigma}, \tilde{\alpha}) \in \mathcal{A}_d^{(p), \text{ex}}(q).$$

By de Jong's description of nonminimal Weierstrass models, relative minimalization recovers a unique relatively-minimal proper regular elliptic surface (X, σ) of exact Faltings height $n \leq d$ and an effective divisor

$$D = \sum_{x \in |\mathbb{P}^1|} e_x x$$

of degree $d - n$, where e_x is the number of prescribed successive blowups above x [dJ02, Section 4.11]. The intrinsic contraction construction in Lemma 5.2 makes $\tilde{\alpha}$ descend uniquely to an automorphism

$$\alpha \in \text{Aut}_{\mathbb{P}^1}(X, \sigma).$$

Because the generic fiber is unchanged, α remains distinct from 1 and $[-1]$. This construction is functorial on arrows and defines a functor

$$\mathcal{A}_d^{(p),\text{ex}}(q) \longrightarrow \bigsqcup_{n=0}^d \bigsqcup_{D \in \text{Sym}^{d-n}(\mathbb{P}^1)(\mathbb{F}_q)} \mathcal{M}_n^{\text{ex}}(q).$$

Conversely, start with an object

$$(X, \sigma, \alpha) \in \mathcal{M}_n^{\text{ex}}(q)$$

and a divisor

$$D \in \text{Sym}^{d-n}(\mathbb{P}^1)(\mathbb{F}_q).$$

Lemma 5.2 constructs the prescribed blowup $(\tilde{X}_D, \tilde{\sigma}_D)$, the unique lift $\tilde{\alpha}_D$, and a coefficient-degree- d Weierstrass model whose minimal desingularization is $(\tilde{X}_D, \tilde{\sigma}_D)$. By the classification of automorphism groups of elliptic curves [Sil09, Appendix A, Proposition 1.2(c)], an origin-preserving automorphism other than 1 and $[-1]$ forces the generic fiber to have $j = 0$ in characteristics 2 and 3. The final assertion of Lemma 5.2 therefore shows that

$$(\tilde{X}_D, \tilde{\sigma}_D, \tilde{\alpha}_D)$$

is an object of $\mathcal{A}_d^{(p),\text{ex}}(q)$.

The lifting and intrinsic contraction constructions in Lemma 5.2 are mutually quasi-inverse on objects and arrows. They recover the support and multiplicities of D and identify both the full automorphism groups and the centralizers of corresponding marked automorphisms. In particular, an arrow forces the divisor D at its source and target to agree, so there are no arrows between summands indexed by distinct divisors. This proves (33).

Taking weighted cardinalities gives

$$\#_q \mathcal{A}_d^{(p),\text{ex}}(q) = \sum_{n=0}^d c_{d-n}(q) \#_q \mathcal{M}_n^{\text{ex}}(q).$$

Using Lemma 5.1 and

$$\#_q \mathcal{M}_n^{\text{ex}}(q) = X_n(q)$$

gives

$$Y_d^{(p)}(q) = \sum_{n=0}^d c_{d-n}(q) X_n(q) = \sum_{e=0}^d c_e(q) X_{d-e}(q),$$

which is (34). ■

Remark 5.4. Proposition 5.3 is an equivalence of essentially finite groupoids over $\mathbb{F}_q$ and the fixed base $\mathbb{P}^1$. It does not assert a family-level equivalence over arbitrary test schemes. In the weighted-projective setting, the quotient and remainder divisors are defined in families with fixed normalized base profile, and the factorization map becomes an isomorphism after stratifying its source and target [BPS22, Lemma 4.32, Proposition 6.1, and Corollary 6.2].

The intrinsic formulation is necessary because the representative-level coefficient division used in the proof of [BPS22, Proposition 6.1] cannot be imposed invariantly on the normalized representatives occurring here. In characteristic three,

a residual translation of the x -coordinate can destroy visible divisibility of C without changing the resolved nonminimal model. Thus, even if

$$h \in H^0(\mathbb{P}^1, \mathcal{O}(d-n))$$

is a section whose zero divisor is D , one cannot require an identity such as

$$C = h^6 C_0$$

in every normalized representative. In characteristic two, writing s and t for the residual translation parameters, the coordinate action mixes B with $As + s^4$ and mixes C with $Bs^2 + s^6 + At + t^2$. In both characteristics, the blowup divisor is intrinsic, whereas visible coefficient divisibility is not.

5.1. Inverting the divisor convolution. The following generating-series argument is the weighted-point-count analogue of the ordinary and inertia factorizations in [BPS22, Lemma 8.6, equations (6)–(7)].

Lemma 5.5. *For every $p \in \{2, 3\}$ and $M \geq 1$,*

$$(35) \quad \sum_{n=0}^M X_n(q) = Y_M^{(p)}(q) - qY_{M-1}^{(p)}(q).$$

Proof. Working with formal power series with rational coefficients, put

$$C(z) = \sum_{e \geq 0} c_e(q) z^e = \frac{1}{(1-z)(1-qz)}, \quad X(z) = \sum_{n \geq 0} X_n(q) z^n,$$

and

$$Y^{(p)}(z) = \sum_{d \geq 0} Y_d^{(p)}(q) z^d.$$

Equation (34) gives

$$Y^{(p)}(z) = C(z)X(z).$$

Consequently,

$$\sum_{M \geq 0} \left(\sum_{n=0}^M X_n(q) \right) z^M = \frac{X(z)}{1-z} = (1-qz)Y^{(p)}(z).$$

Taking the coefficient of z^M proves (35). ■

5.2. The characteristic-three specialization. For $q = 3^r$, the ambient fixed-pair calculation gives

$$(36) \quad Y_d^{(3)}(q) = 2P_d^{(3)}(q) + 2\epsilon_r Q_d^{(3)}(q),$$

where

$$P_d^{(3)}(q) = q^{4d} \frac{q^{2d+1} - 1}{q - 1}$$

and

$$Q_d^{(3)}(q) = \frac{q^{4d+1} - 1}{q - 1}.$$

For every $M \geq 1$, direct subtraction using (2) gives

$$\begin{aligned} P_M^{(3)}(q) - qP_{M-1}^{(3)}(q) &= \alpha_6(q)q^{6M} - \beta_4(q)q^{4M}, \\ Q_M^{(3)}(q) - qQ_{M-1}^{(3)}(q) &= \alpha_4(q)q^{4M} + 1. \end{aligned}$$

Corollary 5.6. *For every $M \geq 1$,*

$$(37) \quad \sum_{n=0}^M X_n(q) = 2(\alpha_6(q)q^{6M} - \beta_4(q)q^{4M}) + 2\epsilon_r(\alpha_4(q)q^{4M} + 1).$$

Proof. By Lemma 5.5 and (36),

$$\sum_{n=0}^M X_n(q) = 2(P_M^{(3)}(q) - qP_{M-1}^{(3)}(q)) + 2\epsilon_r(Q_M^{(3)}(q) - qQ_{M-1}^{(3)}(q)).$$

Substituting the preceding two identities proves the result. ■

5.3. The characteristic-two specialization. For $q = 2^r$, the ambient fixed-pair calculation gives

$$(38) \quad Y_d^{(2)}(q) = P_d^{(2)}(q) + 4\epsilon_r Q_d^{(2)}(q),$$

where

$$P_d^{(2)}(q) = q^{4d} \frac{q^{d+1} - 1}{q - 1}$$

and

$$Q_d^{(2)}(q) = q^{3d} \frac{q^{3d+1} - 1}{q - 1}.$$

For every $M \geq 1$, direct subtraction using (2) gives

$$\begin{aligned} P_M^{(2)}(q) - qP_{M-1}^{(2)}(q) &= \alpha_5(q)q^{5M} - \beta_4(q)q^{4M}, \\ Q_M^{(2)}(q) - qQ_{M-1}^{(2)}(q) &= \alpha_6(q)q^{6M} - \beta_3(q)q^{3M}. \end{aligned}$$

Corollary 5.7. *For every $M \geq 1$,*

$$(39) \quad \sum_{n=0}^M X_n(q) = \alpha_5(q)q^{5M} - \beta_4(q)q^{4M} + 4\epsilon_r(\alpha_6(q)q^{6M} - \beta_3(q)q^{3M}).$$

Proof. By Lemma 5.5 and (38),

$$\sum_{n=0}^M X_n(q) = P_M^{(2)}(q) - qP_{M-1}^{(2)}(q) + 4\epsilon_r(Q_M^{(2)}(q) - qQ_{M-1}^{(2)}(q)).$$

Substituting the preceding two identities proves the result. ■

At $d = 0$ no positive defect is possible, so

$$(40) \quad X_0(q) = Y_0^{(p)}(q) = \begin{cases} 2 + 2\epsilon_r, & q = 3^r, \\ 1 + 4\epsilon_r, & q = 2^r. \end{cases}$$

These give the omitted $M = 0$ values; the characteristic-specific closed-form formulas above require $M \geq 1$.

6. PROOFS OF THE COUNTING FORMULAS

For $M \in \mathbb{Z}_{\geq 0}$, define

$$N_q(M) = \#\{[E/K] : \text{ht}(E) \leq M\}.$$

Thus $N_q(M)$ is the unweighted number of K -isomorphism classes of elliptic curves of Faltings height at most M .

By the definition of the multiplicative discriminant height,

$$N_q(M) = N_{q,\Delta}(q^{12M}).$$

Theorem 6.1. *Let $p \in \{2, 3\}$ and $r \geq 1$, and put $q = p^r$. For every $M \in \mathbb{Z}_{\geq 1}$, the following formulas hold.*

If $p = 3$, then

$$\begin{aligned} N_q(M) = & 2 \left(\frac{q^9 - 1}{q^8 - q^7} q^{10M} - q^{2M} - \beta_4(q) q^{4M} - 1 \right) \\ & + 2(\alpha_6(q) q^{6M} - \beta_4(q) q^{4M}) + 2\epsilon_r(\alpha_4(q) q^{4M} + 1). \end{aligned}$$

If $p = 2$, then

$$\begin{aligned} N_q(M) = & 2 \left(\frac{q^9 - 1}{q^8 - q^7} q^{10M} - q^{2M} - \beta_6(q) q^{6M} \right) \\ & + \alpha_5(q) q^{5M} - \beta_4(q) q^{4M} + 4\epsilon_r(\alpha_6(q) q^{6M} - \beta_3(q) q^{3M}). \end{aligned}$$

For $M = 0$,

$$N_q(0) = \begin{cases} 2q + 2 + 2\epsilon_r, & p = 3, \\ 2q + 1 + 4\epsilon_r, & p = 2. \end{cases}$$

Equivalently,

$$N_q(0) = \begin{cases} 2q + 2, & p = 3 \text{ and } r \text{ odd}, \\ 2q + 4, & p = 3 \text{ and } r \text{ even}, \\ 2q + 1, & p = 2 \text{ and } r \text{ odd}, \\ 2q + 5, & p = 2 \text{ and } r \text{ even}. \end{cases}$$

Proof. Since the Faltings height is a nonnegative integer and $U_n(q)$ counts the curves of exact height n , for every $M \in \mathbb{Z}_{\geq 0}$ one has

$$N_q(M) = \sum_{n=0}^M U_n(q).$$

For $M \geq 1$, summing the marked-inertia identity

$$U_n(q) = 2I_n(q) + X_n(q)$$

of Lemma 2.1 therefore gives

$$N_q(M) = 2 \sum_{n=0}^M I_n(q) + \sum_{n=0}^M X_n(q).$$

Proposition 2.5 gives

$$\sum_{n=0}^M I_n(q) = \frac{q^9 - 1}{q^8 - q^7} q^{10M} - q^{2M} - \beta_4(q) q^{4M} - 1 \quad (p = 3),$$

and

$$\sum_{n=0}^M I_n(q) = \frac{q^9 - 1}{q^8 - q^7} q^{10M} - q^{2M} - \beta_6(q) q^{6M} \quad (p = 2).$$

If $q = 3^r$, Corollary 5.6 gives

$$\sum_{n=0}^M X_n(q) = 2(\alpha_6(q) q^{6M} - \beta_4(q) q^{4M}) + 2\epsilon_r(\alpha_4(q) q^{4M} + 1).$$

If $q = 2^r$, Corollary 5.7 gives

$$\sum_{n=0}^M X_n(q) = \alpha_5(q) q^{5M} - \beta_4(q) q^{4M} + 4\epsilon_r(\alpha_6(q) q^{6M} - \beta_3(q) q^{3M}).$$

Substitution gives the two formulas asserted in the theorem.

For $M = 0$, Proposition 2.5 gives $I_0(q) = q$, while (40) gives the corresponding value of $X_0(q)$. The marked-inertia identity therefore yields

$$N_q(0) = U_0(q) = \begin{cases} 2q + 2 + 2\epsilon_r, & q = 3^r, \\ 2q + 1 + 4\epsilon_r, & q = 2^r. \end{cases}$$

The equivalent four-case formulation follows from the definition of ϵ_r . ■

Proof of Theorem 1.1. For arbitrary real $B \geq 1$, let B^* be the largest element of $q^{12\mathbb{Z}_{\geq 0}}$ not exceeding B . Since

$$H_{\Delta}(E) = q^{12\text{ht}(E)}$$

and $\text{ht}(E) \in \mathbb{Z}_{\geq 0}$, it follows that

$$H_{\Delta}(E) \leq B \iff H_{\Delta}(E) \leq B^*.$$

Consequently,

$$N_{q,\Delta}(B) = N_{q,\Delta}(B^*).$$

If $B^* = 1$, then

$$N_{q,\Delta}(B) = N_{q,\Delta}(1) = N_q(0),$$

so the height-zero formula follows from Theorem 6.1.

Suppose that $B^* > 1$. Write

$$B^* = q^{12M} \quad (M \geq 1).$$

Then

$$\begin{aligned} q^{10M} &= (B^*)^{5/6}, & q^{6M} &= (B^*)^{1/2}, & q^{5M} &= (B^*)^{5/12}, \\ q^{4M} &= (B^*)^{1/3}, & q^{3M} &= (B^*)^{1/4}, & q^{2M} &= (B^*)^{1/6}. \end{aligned}$$

Substituting these identities into Theorem 6.1, and using

$$N_{q,\Delta}(B) = N_{q,\Delta}(B^*) = N_q(M),$$

gives the two positive-height formulas asserted in Theorem 1.1. ■

7. TWISTS AND THE LOWER-ORDER TERMS

7.1. Finite-field twists at height zero. By an $\mathbb{F}_q$ -form, or *twist*, of an elliptic curve $E/\overline{\mathbb{F}}_q$ we mean an elliptic curve $E'/\mathbb{F}_q$ such that

$$E'_{\overline{\mathbb{F}}_q} \simeq E.$$

Kronberg–Soomro–Top determine the finite-field forms of elliptic curves in characteristics 2 and 3. Their results give

$$\# \left\{ \begin{array}{l} \mathbb{F}_q\text{-isomorphism classes} \\ \text{of elliptic curves with } j = 0 \end{array} \right\} = \begin{cases} 4 + 2\epsilon_r, & q = 3^r, \\ 3 + 4\epsilon_r, & q = 2^r, \end{cases}$$

where the characteristic-three statement is [KST17, Proposition 2.2] and the characteristic-two statement is [KST17, Proposition 3.2].

Write

$$I_0^{j=0}(q), \quad X_0^{j=0}(q), \quad U_0^{j=0}(q)$$

for the weighted, marked-inertia, and unweighted height-zero counts restricted to the locus $j = 0$.

Every height-zero elliptic curve over $K = \mathbb{F}_q(t)$ is constant. Indeed, its fundamental line bundle has degree zero and hence is isomorphic to $\mathcal{O}_{\mathbb{P}^1}$. A global minimal Weierstrass equation therefore has coefficients in $H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}) = \mathbb{F}_q$, so its generic fiber is obtained by base change from $\mathbb{F}_q$.

At coefficient degree $d = 0$, no positive minimality defect is possible. Setting $d = 0$ in (10) or (20) therefore gives

$$I_0^{j=0}(q) = 1.$$

By the classification of origin-preserving automorphism groups of elliptic curves [Sil09, Appendix A, Proposition 1.2(c)], automorphisms other than 1 and $[-1]$ occur only at $j = 0$ in characteristics 2 and 3. Hence

$$X_0^{j=0}(q) = X_0(q).$$

Equation (40) then gives

$$X_0^{j=0}(q) = \begin{cases} 2 + 2\epsilon_r, & q = 3^r, \\ 1 + 4\epsilon_r, & q = 2^r. \end{cases}$$

Consequently, the marked-inertia identity gives

$$U_0^{j=0}(q) = 2I_0^{j=0}(q) + X_0^{j=0}(q) = \begin{cases} 4 + 2\epsilon_r, & q = 3^r, \\ 3 + 4\epsilon_r, & q = 2^r. \end{cases}$$

Thus the height-zero calculation recovers the twist numbers of Kronberg–Soomro–Top.

For each $j \in \mathbb{F}_q^\times$, there are exactly two $\mathbb{F}_q$ -isomorphism classes of elliptic curves with j -invariant j [KST17, Propositions 2.2 and 3.2]. Hence the total height-zero count is

$$2(q-1) + (4 + 2\epsilon_r) = 2q + 2 + 2\epsilon_r$$

in characteristic three, and

$$2(q-1) + (3 + 4\epsilon_r) = 2q + 1 + 4\epsilon_r$$

in characteristic two, as asserted in Theorem 6.1.

7.2. Twisted Galois invariants in positive height. The finite-field twist numbers do not give a uniform multiplicative factor on the positive-height $j = 0$ locus.

Fix a separable closure K^{sep} of K containing $\overline{\mathbb{F}}_q$, and write

$$G_{\mathbb{F}_q} = \text{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q), \quad G_K = \text{Gal}(K^{\text{sep}}/K).$$

The use of a separable closure is important here because $K = \mathbb{F}_q(t)$ is imperfect.

By [dJ02, Lemma 10.1], every elliptic curve over K with constant j -invariant becomes, over a finite separable extension of K , isomorphic to the base change of an elliptic curve over $\mathbb{F}_q$. For the locus considered here, fix a supersingular elliptic curve $E_0/\mathbb{F}_q$ with $j(E_0) = 0$. Let

$$\mathcal{A}_0 = \underline{\text{Aut}}_{\mathbb{F}_q}(E_0, O), \quad \Gamma_0 = \mathcal{A}_0(\overline{\mathbb{F}}_q) \simeq \text{Aut}(E_{0,K^{\text{sep}}}, O).$$

Both $G_{\mathbb{F}_q}$ and G_K act continuously on Γ_0 through their actions on $\overline{\mathbb{F}}_q$. For τ in either group and $g \in \Gamma_0$, we denote by $\tau(g)$ the automorphism obtained by applying τ to the coefficients of g .

By Remark 2.2, the relevant nonempty Isom schemes are torsors under the corresponding base changes of the finite étale group scheme $\mathcal{A}_0$. Hence forms are classified by continuous nonabelian Galois cohomology even over the imperfect field K .

For a profinite group G acting continuously on Γ_0 , the pointed set $H^1(G, \Gamma_0)$ consists of continuous maps

$$\xi: G \longrightarrow \Gamma_0$$

satisfying

$$\xi_{\sigma\tau} = \xi_\sigma \sigma(\xi_\tau),$$

modulo the equivalence relation

$$\xi'_\tau = a \xi_\tau \tau(a)^{-1}, \quad a \in \Gamma_0.$$

The $\mathbb{F}_q$ -forms of $E_{0,\overline{\mathbb{F}}_q}$ are classified by

$$H^1(G_{\mathbb{F}_q}, \Gamma_0),$$

whereas the K -forms of $E_{0,K^{\text{sep}}}$ are classified by

$$H^1(G_K, \Gamma_0).$$

Because $K = \mathbb{F}_q(t)$ is regular over $\mathbb{F}_q$, restriction to $\overline{\mathbb{F}}_q$ is surjective and gives the exact sequence

$$1 \longrightarrow \text{Gal}(K^{\text{sep}}/\overline{\mathbb{F}}_q(t)) \longrightarrow G_K \longrightarrow G_{\mathbb{F}_q} \longrightarrow 1.$$

Since every element of Γ_0 is defined over $\overline{\mathbb{F}}_q$, the kernel acts trivially on Γ_0 . Consequently, the action of G_K factors through $G_{\mathbb{F}_q}$.

Precomposition of cocycles with the displayed surjection defines the inflation map

$$H^1(G_{\mathbb{F}_q}, \Gamma_0) \longrightarrow H^1(G_K, \Gamma_0).$$

Its image consists exactly of the classes obtained by base change from $\mathbb{F}_q$. Thus a nonconstant isotrivial form corresponds to a class that is not inflated from $G_{\mathbb{F}_q}$.

Put

$$\mathcal{A}_{0,K} = \mathcal{A}_0 \times_{\mathbb{F}_q} \text{Spec } K, \quad \mathcal{A}_{0,\mathbb{P}^1} = \mathcal{A}_0 \times_{\mathbb{F}_q} \mathbb{P}_{\mathbb{F}_q}^1.$$

Call a class in $H^1(G_K, \Gamma_0)$ *unramified at a closed point x* if its corresponding $\mathcal{A}_{0,K}$ -torsor extends to an $\mathcal{A}_{0,\mathbb{P}^1}$ -torsor over

$$\text{Spec } \mathcal{O}_{\mathbb{P}^1, x}.$$

An everywhere-unramified torsor extends uniquely over $\mathbb{P}_{\mathbb{F}_q}^1$ [Sta18, Tag 0BTF]. Moreover,

$$H_{\text{ét}}^1(\mathbb{P}_{\mathbb{F}_q}^1, \mathcal{A}_{0,\mathbb{P}^1}) \simeq H^1(G_{\mathbb{F}_q}, \Gamma_0),$$

because

$$\pi_1^{\text{ét}}(\mathbb{P}_{\mathbb{F}_q}^1) = 1.$$

Under this identification, restriction to the generic point is the inflation map displayed above. Thus every everywhere-unramified class is inflated from $G_{\mathbb{F}_q}$ and defines a constant form. Consequently, a nonconstant isotrivial form must be ramified at some place of $\mathbb{P}^1$.

Since constant elliptic curves have height zero, every positive-height curve on the $j = 0$ locus is a nonconstant isotrivial form and is therefore ramified somewhere.

For a K -form E of E_0 , choose an isomorphism

$$\psi: E_{K^{\text{sep}}} \xrightarrow{\sim} E_{0,K^{\text{sep}}},$$

and put

$$\xi_\tau = \psi \circ \tau(\psi)^{-1} \in \Gamma_0.$$

Under ψ , an automorphism of $E_{K^{\text{sep}}}$ corresponds to an element $g \in \Gamma_0$. It is defined over K precisely when it is fixed by the descent datum, which means

$$\xi_\tau \tau(g) \xi_\tau^{-1} = g \quad \text{for every } \tau \in G_K.$$

Thus

$$\text{Aut}_K(E, O) \simeq \{g \in \Gamma_0 : \xi_\tau \tau(g) \xi_\tau^{-1} = g \text{ for every } \tau \in G_K\}.$$

Equivalently,

$$g \xi_\tau = \xi_\tau \tau(g) \quad \text{for every } \tau \in G_K.$$

This fixed group can depend on the cocycle ξ . Lemmas 3.2 and 4.2 classify it directly, while Propositions 3.3 and 4.4 count the corresponding marked pairs with their centralizer weights. Consequently, the finite-field twist count cannot be used as a uniform multiplier in positive height.

7.3. The characteristic-three lower-order terms. Suppose that $q = 3^r$. For $M \geq 1$, put $B = q^{12M}$.

The dominant marked fixed-pair locus is

$$A = -s^2, \quad s \neq 0.$$

The sublocus with $u = 1$ supports the order-three automorphisms

$$(x, y) \longmapsto (x + s, y),$$

while the sublocus with $u = -1$ consists of their products with inversion and supports automorphisms of order six. Together, the subloci with $u = 1$ and $u = -1$

have ambient marked count $2P_d^{(3)}(q)$, of leading growth q^{6d} . After inversion of the defect recursion, their combined contribution is

$$2(\alpha_6(q)q^{6M} - \beta_4(q)q^{4M}) = 2\alpha_6(q)B^{1/2} - 2\beta_4(q)B^{1/3}.$$

The second family of marked fixed-pair loci is cut out by

$$u^2 = -1.$$

Their union is defined over $\mathbb{F}_q$ for every r . Over $\overline{\mathbb{F}}_q$, it is the disjoint union of two fixed-pair loci, one for each root of $u^2 = -1$. These two loci are individually defined over $\mathbb{F}_q$ exactly when r is even. When r is odd, Frobenius exchanges them, so their union has no $\mathbb{F}_q$ -points. Indeed, an $\mathbb{F}_q$ -point fixed by Frobenius would have to lie in both disjoint geometric components. The corresponding marked automorphisms have order four. Their contribution is

$$2\epsilon_r(\alpha_4(q)q^{4M} + 1) = 2\epsilon_r\alpha_4(q)B^{1/3} + 2\epsilon_r.$$

The negative terms of order $B^{1/3}$ have two distinct origins. The summand

$$-2\beta_4(q)B^{1/3}$$

coming from the marked loci is not the count of a negative geometric stratum. It is created when the minimality defect is removed from the positive ambient marked count. When r is odd, it is the entire extra-inertia contribution of order $B^{1/3}$.

A second summand

$$-2\beta_4(q)B^{1/3} - 2$$

comes from the corrected weighted count, namely from the additional nonsmooth equations in the quasi-elliptic-type stratum of Lemma 2.3. Together these contributions explain the coefficient

$$2(\epsilon_r\alpha_4(q) - 2\beta_4(q))$$

and the constant term $2\epsilon_r - 2$ in Theorem 1.1.

The characteristic-three cubic-twist family

$$y^2 = x^3 - x + g(t), \quad g(t) \in K,$$

lies on the wild fixed-pair locus $A = -s^2$; on the standard affine chart one may take $s = \pm 1$. Indeed, after adjoining an element a satisfying

$$a^3 - a = -g(t),$$

the change of variables $x = X + a$ gives the constant equation

$$y^2 = X^3 - X.$$

These are the characteristic-three examples underlying the vector-bundle obstruction studied by Landesman [Lan24]. The present argument does not require a weighted-projective presentation of $\overline{\mathcal{M}}_{1,1}$. Instead, it uses finite groupoids of Weierstrass equations and de Jong's invariant blowup divisor, both of which remain available in characteristic three.

7.4. The characteristic-two lower-order terms. Suppose that $q = 2^r$. For $M \geq 1$, put $B = q^{12M}$. Over an algebraic closure, the origin-preserving automorphism group of a $j = 0$ elliptic curve in characteristic two has order 24. Its order-four elements form one conjugacy class of six elements, with centralizer of order four. Its order-three elements form two conjugacy classes of four elements each, and its order-six elements likewise form two conjugacy classes of four elements each; their centralizers have order six [KST17, Proposition 3.1].

Over $\overline{\mathbb{F}}_q$, the geometric type-(II) fixed pairs are partitioned into four loci indexed by

$$u \in \mu_3(\overline{\mathbb{F}}_q) \setminus \{1\}, \quad \varepsilon \in \{0, 1\}.$$

These four loci are individually defined over $\mathbb{F}_q$ exactly when the two nontrivial cube roots of unity lie in $\mathbb{F}_q$, that is, when

$$\mu_3(\overline{\mathbb{F}}_q) \subset \mathbb{F}_q^\times.$$

Equivalently,

$$\mathbb{F}_4 \subset \mathbb{F}_q,$$

or r is even. When r is odd, Frobenius exchanges the two loci indexed by the nontrivial cube roots of unity for each value of ε . Since the exchanged loci are disjoint, the type-(II) incidence has no $\mathbb{F}_q$ -points.

The type-(I) locus supports the order-four automorphisms whose square is inversion. Its contribution after removal of the minimality defect is

$$\alpha_5(q)q^{5M} - \beta_4(q)q^{4M} = \alpha_5(q)B^{5/12} - \beta_4(q)B^{1/3}.$$

When r is even, the four type-(II) fixed-pair loci support the order-three and order-six automorphisms. Their combined contribution is

$$4\epsilon_r(\alpha_6(q)q^{6M} - \beta_3(q)q^{3M}) = 4\epsilon_r\alpha_6(q)B^{1/2} - 4\epsilon_r\beta_3(q)B^{1/4}.$$

The additional nonsmooth equations in the quasi-elliptic-type stratum of Lemma 2.3 contribute

$$-2\beta_6(q)B^{1/2}$$

through the corrected weighted count. Thus the total coefficient of $B^{1/2}$ is

$$2(2\epsilon_r\alpha_6(q) - \beta_6(q)).$$

When r is odd, this coefficient is the negative quantity $-2\beta_6(q)$.

As in characteristic three, the negative terms

$$-\beta_4(q)B^{1/3} \quad \text{and} \quad -4\epsilon_r\beta_3(q)B^{1/4}$$

do not represent negative geometric strata. They are the lower terms created by removing nonminimal models from positive ambient fixed-pair counts.

7.5. Features common to both characteristics. In both characteristics, the term

$$-2B^{1/6}$$

is the term $-2q^{2M}$ in twice the corrected cumulative weighted count. Together with the characteristic-specific analyses above, this accounts for every term in Theorem 1.1 and completes the geometric interpretation of the exact counting formulas.

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Jun-Yong Park – june.park@sydney.edu.au

SCHOOL OF MATHEMATICS AND STATISTICS, UNIVERSITY OF SYDNEY, AUSTRALIA